

Engineering Mathematics

Previous Year Questions • 1988–2026

PROBABILITY AND STATISTICS

(EC, EE, IN, CS, ME, CE, CH, PI)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Probability and Statistics (EC/EE/IN/CS/ME/CE/CH/PI), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- AIR 27 — GATE (ECE)
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- 5+ years of teaching experience



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- 3+ years in teaching & the VLSI industry



> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:





Easy



Medium



Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Branch Snapshot

A Maths unit can span up to eight branches (EC, EE, IN, CS, ME, CE, CH, PI), each weighted very differently in the real exam. Right after every Unit divider, a **Branch Snapshot** page lists every branch that unit covers, in the same order the questions themselves follow, each in its own compact box:

Questions	the real total for that branch.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy branch showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Branch Snapshot*

Branch: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real EC questions on this unit; on average EC has drawn about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** for your branch points to a consistently high-yield topic worth prioritizing; a low one is largely someone else's problem. A wide **Range** usually means the topic's importance has shifted over the years, so it is worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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1.	Probability and Statistics	1

UNIT

I

Probability and Statistics

Topics

1. Basic Probability - Definitions and Axioms
2. Conditional Probability and Baye's Theorem
3. Random Variables - Discrete and Continuous
4. Exception, Variance and Standard Deviation
5. Binomial and Poisson Distribution
6. Uniform and Exponential Distribution
7. Gaussian (Normal) Distribution
8. Multivariable Random Variables
9. Correlation and Regression

Branches: EC,EE,IN,CS,ME,CE,CH,PI

Branch Snapshots*

EC — Electronics & Communication Engineering

Questions: 53 **MCQ:** 29 **MSQ:** 1 **NAT:** 23
1M: 23 **2M:** 29 **Desc:** 1 **Avg Weightage** ~3M **Range:** 1M(2022)–4M(2013)

EE — Electrical Engineering

Questions: 34 **MCQ:** 23 **MSQ:** 1 **NAT:** 10
1M: 16 **2M:** 18 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2013)–3M(2025)

IN — Instrumentation Engineering

Questions: 33 **MCQ:** 21 **MSQ:** 0 **NAT:** 12
1M: 16 **2M:** 17 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2022)–4M(2018)

CS — Computer Science & IT

Questions: 58 **MCQ:** 31 **MSQ:** 2 **NAT:** 25
1M: 25 **2M:** 33 **Desc:** 0 **Avg Weightage** ~4M **Range:** 1M(2013)–5M(2026)

ME — Mechanical Engineering

Questions: 67 **MCQ:** 50 **MSQ:** 0 **NAT:** 17
1M: 29 **2M:** 38 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2025)–5M(2013)

CE — Civil Engineering

Questions: 58 **MCQ:** 32 **MSQ:** 1 **NAT:** 25
1M: 31 **2M:** 27 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2020)–6M(2016)

CH — Chemical Engineering

Questions: 26 **MCQ:** 18 **MSQ:** 1 **NAT:** 7
1M: 11 **2M:** 15 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2026)–4M(2014)

PI — Production & Industrial Engineering

Questions: 42 **MCQ:** 29 **MSQ:** 0 **NAT:** 13
1M: 20 **2M:** 22 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2025)–5M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Electronics & Communication Engineering

Q3.26.1 MCQ 2026 2M

Let X, N, Y and Z be random variables. The variables X and N are independent of each other. X is uniformly distributed between -1 and 1; N follows Normal distribution with zero mean and unity variance. Y and Z are defined as, $Y = X+N$ and $Z = X^2+N$. Which of the following pairs represents the values of the cross-correlations $R_{XY} = E[XY]$ and $R_{XZ} = E[XZ]$ (without normalization)?

A. $\frac{1}{3}$ and 0 **B.** $\frac{1}{3}$ and $\frac{1}{9}$

C. $\frac{1}{3}$ and $\frac{1}{3}$ **D.** 1 and 0

Q3.25.1 MCQ 2025 1M

A pot contains two red balls and two blue balls. Two balls are drawn from this pot randomly without replacement. What is the probability that the two balls drawn have different colours?

A. $\frac{2}{3}$ **B.** $\frac{1}{3}$ **C.** $\frac{1}{2}$ **D.** 1

Q3.25.2 NAT 2025 2M

Two fair dice (with faces labeled 1, 2, 3, 4, 5, and 6) are rolled. Let the random variable X denote the sum of the outcomes obtained. The expectation of X is _____ *(rounded off to two decimal places).*

Q3.24.1 NAT 2024 2M

Suppose X and Y are independent and identically distributed random variables that are distributed uniformly in the interval [0,1]. The probability that $X \geq Y$ is _____.

Q3.22.1 NAT 2022 1M

The bar graph shows the frequency of the number of wickets taken in a match by a bowler in her career. For example, in 17 of her matches, the bowler has taken 5 wickets each. The median number of wickets taken by the bowler in a match is _____. *(rounded off to one decimal place).*

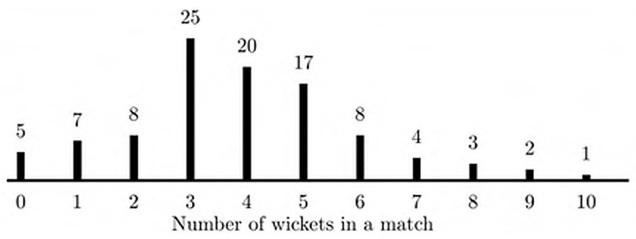


Fig. Q3.22.1

Q3.21.1 MCQ 2021 1M

Two continuous random variables X and Y are related as

$$Y = 2X + 3$$

Let σ_X^2 and σ_Y^2 denote the variances of X and Y, respectively. The variances are related as

A. $\sigma_Y^2 = 2\,\sigma_X^2$ **B.** $\sigma_Y^2 = 4\,\sigma_X^2$

C. $\sigma_Y^2 = 5\,\sigma_X^2$ **D.** $\sigma_Y^2 = 25\,\sigma_X^2$

Q3.21.2 MCQ 2021 2M

A box contains the following three coins.

I. A fair coin with head on one face and tail on the other face.

II. A coin with heads on both the faces.

III. A coin with tails on both the faces.

A coin is picked randomly from the box and tossed. Out of the two remaining coins in the box, one coin is then picked randomly and tossed. If the first toss results in a head, the probability of getting a head in the second toss is

A. $\frac{2}{5}$ **B.** $\frac{1}{3}$ **C.** $\frac{1}{2}$ **D.** $\frac{2}{3}$

Q3.20.1 NAT 2020 1M

The two sides of a fair coin are labelled as 0 and 1. The coin is tossed two times independently. Let M and N denote the labels corresponding to the outcomes of those tosses. For a random variable X, defined as $X = min(M, N)$, the expected value of $E(X)$ **(rounded off to two decimal places)** is _____.

Q3.20.2 NAT 2020 2M

X is a random variable with uniform probability density function in the interval $[-2, 10]$. For $Y = 2X - 6$, the conditional probability $P(Y \leq 7 | X \geq 5)$ **(rounded off to three decimal places)** is _____.

Q3.19.1 NAT 2019 1M

If X and Y are random variables such that $E[2X + Y] = 0$ and $E[X + 2Y] = 33$, then $E[X] + E[Y]$ = _____.

Q3.19.2 NAT 2019 1M

Let Z be an exponential random variable with mean 1. That is, the cumulative distribution function of Z is given by

$$F_Z(x) = \begin{cases} 1 - e^{-x} & if\ x \geq 0 \\ 0 & if\ x < 0 \end{cases}$$

Then $Pr(Z > 2|Z > 1)$, rounded off to two decimal places, is equal to _____.

Q3.18.1 NAT 2018 1M

Let X_1, X_2, X_3 and X_4 be independent normal random variables with zero mean and unit variance. The probability that X_4 is the smallest among the four is _____.

Q3.17.1 NAT 2017 1M

Three fair cubical dice are thrown simultaneously. The probability that all three dice have the same number of dots on the faces showing up is (up to third decimal place) _____.

Q3.17.2 NAT 2017 2M

Passengers try repeatedly to get a seat reservation in any train running between two stations until they are successful. If there is 40% chance of getting reservation in any attempt by a passenger. Then the average number of attempts that passengers need to get a seat reserved is _____.

Q3.16.1 NAT 2016 1M

The second moment of a Poisson-distributed random variable is 2. The mean of the random variable is _____

Q3.16.2 NAT 2016 2M

Two random variables X and Y are distributed according to

$$f_{X,Y}(x,y) = \begin{cases} (x+y), & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

The probability $P(X + Y \le 1)$ is _____.

Q3.16.3 NAT 2016 1M

The probability of getting a "head" in a single toss of a biased coin is 0.3. The coin is tossed repeatedly till a "head" is obtained. If the tosses are independent, then the probability of getting "head" for the first time in the fifth toss is _____

Q3.15.1 MCQ 2015 1M

Suppose A and B are two independent events with probabilities $P(A) \neq 0$ and $P(B) \neq 0$. Let $\overline{A}$ and $\overline{B}$ be their complements. Which one of the following statements is FALSE?

A. $P(A \cap B) = P(A)P(B)$

B. $P(A|B) = P(A)$

C. $P(A \cup B) = P(A) + P(B)$

D. $P(\overline{A} \cap \overline{B}) = P(\overline{A})P(\overline{B})$

Q3.15.2 NAT 2015 2M

Let the random variable X represent the number of times a fair coin needs to be tossed till two consecutive heads appear for the first time. The expectation of X is _____

Q3.15.3 MCQ 2015 2M

Let $X \in \{0, 1\}$ and $Y \in \{0, 1\}$ be two independent binary random variables. If $P(X = 0) = p$ and $P(Y = 0) = q$, then $P(X + Y \ge 1)$ is equal to

A. $pq + (1 - p)(1 - q)$ **B.** pq

C. $p(1 - q)$ **D.** $1 - pq$

Q3.15.4 NAT 2015 2M

A fair die with faces $\{1, 2, 3, 4, 5, 6\}$ is thrown repeatedly till '3' is observed for the first time. Let X denote the number of times the die is thrown. The expected value of X is _____.

Q3.15.5 NAT 2015 2M

The variance of the random variable X with probability density function $f(x) = \frac{1}{2}|x|e^{-|x|}$ is _____

Q3.14.1 NAT 2014 1M

In a housing society, half of the families have a single child per family, while the remaining half have two children per family. The probability that a child picked at random, has a sibling is _____

Q3.14.2 MCQ 2014 2M

Let X be a real-valued random variable with $E[X]$ and $E[X^2]$ denoting the mean values of X and X^2 respectively. The relation which always holds true is

A. $(E[X])^2 > E[X^2]$ **B.** $E[X^2] \ge (E[X])^2$

C. $E[X^2] = (E[X])^2$ **D.** $E[X^2] > (E[X])^2$

Q3.14.3 NAT 2014 1M

Let X be a random variable which is uniformly chosen from the set of positive numbers less than 100. The expectation, $E[X]$, is _____

Q3.14.4 MCQ 2014 1M

An unbiased coin is tossed an infinite number of times. The probability that the fourth head appears at the tenth toss is

A. 0.067 **B.** 0.073 **C.** 0.082 **D.** 0.091

Q3.14.5 NAT 2014 2M

A fair coin is tossed repeatedly till both head and tail appear at least once. The average number of tosses required is _____

Q3.14.6 NAT 2014 2M

Let X_1, X_2 and X_3 be independent and identically distributed random variables with the uniform distribution on $[0, 1]$. The probability that $P\{X_1 + X_2 \leq X_3\}$ is _____.

Q3.14.7 NAT 2014 1M

Let X be a zero mean unit variance Gaussian random variable. $E[|X|]$ is equal to _____

Q3.14.8 MCQ 2014 1M

If calls arrive at a telephone exchange such that the time of arrival of any call is independent of the time of arrival of earlier or future calls, the probability distribution function of the total number of calls in a fixed time interval will be

A. Poisson **B.** Gaussian

C. Exponential **D.** Gamma

Q3.14.9 NAT 2014 2M

Parcels from sender S to receiver R pass sequentially through two post-offices. Each post-office has a probability $\frac{1}{5}$ of losing an incoming parcel, independently of all other parcels. Given that a parcel is lost, the probability that it was lost by the second post-office is _____

Q3.13.1 MCQ 2013 2M

Let U and V be two independent zero mean Gaussian random variables of variances $\frac{1}{4}$ and $\frac{1}{9}$ respectively. The probability $P(3V \geq 2U)$ is

A. $\frac{4}{9}$ **B.** $\frac{1}{2}$ **C.** $\frac{2}{3}$ **D.** $\frac{5}{9}$

Q3.13.2 MCQ 2013 2M

Consider two identically distributed zero-mean random variables U and V. Let the cumulative distribution functions of U and $2V$ be $F(x)$ and $G(x)$ respectively. Then, for all values of x

A. $F(x) - G(x) \leq 0$ **B.** $F(x) - G(x) \geq 0$

C. $(F(x) - G(x)) \cdot x \leq 0$ **D.** $(F(x) - G(x)) \cdot x \geq 0$

Q3.12.1 MCQ 2012 1M

Two independent random variables X and Y are uniformly distributed in the interval $[-1, 1]$. The probability that $\max[X, Y]$ is less than $\frac{1}{2}$ is

A. $\frac{3}{4}$ **B.** $\frac{9}{16}$ **C.** $\frac{1}{4}$ **D.** $\frac{2}{3}$

Q3.12.2 MCQ 2012 2M

A fair coin is tossed till a head appears for the first time. The probability that the number of required tosses is odd, is

A. $\frac{1}{3}$ **B.** $\frac{1}{2}$ **C.** $\frac{2}{3}$ **D.** $\frac{3}{4}$

Q3.11.1 MCQ 2011 2M

A fair dice is tossed two times. The probability that the second toss results in a value that is higher than the first toss is

A. $\frac{2}{36}$ **B.** $\frac{2}{6}$ **C.** $\frac{5}{12}$ **D.** $\frac{1}{2}$

Q3.10.1 MCQ 2010 2M

A fair coin is tossed independently four times. The probability of the event "the number of times heads show up is more than the number of times tails show up" is

A. $\frac{1}{16}$ **B.** $\frac{1}{8}$ **C.** $\frac{1}{4}$ **D.** $\frac{5}{16}$

Q3.09.1 MCQ 2009 1M

A fair coin is tossed 10 times. What is the probability that ONLY the first two tosses will yield heads?

A. $\left(\frac{1}{2}\right)^2$ **B.** ${}^{10}C_2\left(\frac{1}{2}\right)^2$ **C.** $\left(\frac{1}{2}\right)^{10}$ **D.** ${}^{10}C_2\left(\frac{1}{2}\right)^{10}$

Q3.09.2 MCQ 2009 2M

Consider two independent random variables X and Y with identical distributions. The variables X and Y take values 0, 1 and 2 with probabilities $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{4}$ respectively. What is the conditional probability $P(X + Y = 2 | X - Y = 0)$?

A. 0 **B.** $\frac{1}{16}$ **C.** $\frac{1}{6}$ **D.** 1

Q3.09.3 MCQ 2009 2M

A discrete random variable X takes values from 1 to 5 with probabilities as shown in the table. A student calculates the mean of X as 3.5 and her teacher calculates the variance of X as 1.5. Which of the following statements is true?

k	1	2	3	4	5
$P(X=k)$	0.1	0.2	0.4	0.2	0.1

A. Both the student and the teacher are right

B. Both the student and the teacher are wrong

C. The student is wrong but the teacher is right

D. The student is right but the teacher is wrong

Q3.08.1 MCQ 2008 2M

$P_X(x) = Me^{-2|x|} + Ne^{-3|x|}$ is the probability density function for the real random variable X, over the entire x axis. M and N are both positive real numbers. The equation relating M and N is

A. $M + \frac{2}{3}N = 1$ **B.** $2M + \frac{1}{3}N = 1$

C. $M + N = 1$ **D.** $M + N = 3$

Q3.07.1 MCQ 2007 1M

If E denotes expectation, the variance of a random variable X is given by

A. $E[X^2] - E^2[X]$ **B.** $E[X^2] + E^2[X]$

C. $E[X^2]$ **D.** $E^2[X]$

Q3.07.2 MCQ 2007 2M

An examination consists of two papers, Paper 1 and Paper 2. The probability of failing in Paper 1 is 0.3 and that in Paper 2 is 0.2. Given that a student has failed in Paper 2, the probability of failing in Paper 1 is 0.6. The probability of a student failing in both the papers is

A. 0.5 **B.** 0.18 **C.** 0.12 **D.** 0.06

Q3.06.1 MCQ 2006 1M

A probability density function is of the form

$$p(x) = Ke^{-\alpha|x|}, x \in (-\infty, \infty)$$

The value of K is

A. 0.5 **B.** 1 **C.** 0.5α **D.** α

Q3.06.2 MCQ 2006 2M

Three companies X, Y, and Z supply computers to a university. The percentage of computers supplied by them and the probability of those being defective are tabulated below

Company	*% of Computer Supplied*	*Probability of being supplied defective*
X	60%	0.01
Y	30%	0.02
Z	10%	0.03

Given that a computer is defective, the probability that was supplied by Y is

A. 0.1 **B.** 0.2 **C.** 0.3 **D.** 0.4

Q3.06.3 MCQ 2006 2M

A uniformly distributed random variable X with probability density function

$$f_x(x) = \frac{1}{10}[u(x+5) - u(x-5)]$$

where $u(\cdot)$ is the unit step function is passed through a transformation given in the figure below. The probability density function of the transformed random variable Y would be

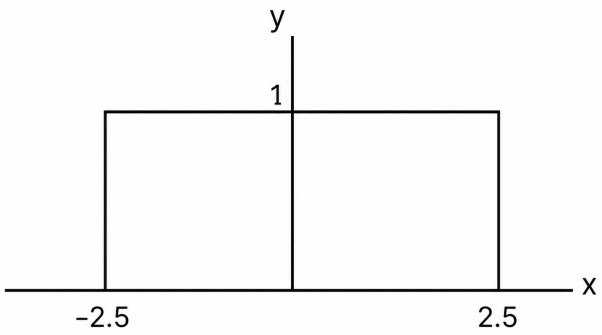


Fig. Q3.06.3

A. $f_y(y) = \frac{1}{5}[u(y+2.5) - u(y-2.5)]$

B. $f_y(y) = 0.5\delta(y) + 0.5\delta(y-1)$

C. $f_y(y) = 0.25\delta(y+2.5) + 0.25\delta(y-2.5) + 0.5\delta(y)$

D. $f_y(y) = 0.25\delta(y+2.5) + 0.25\delta(y-2.5)] + \frac{1}{10}[u(y+2.5) - u(y-2.5)$

Q3.05.1 MCQ 2005 1M

A fair dice is rolled twice. The probability that an odd number will follow an even number is

A. $\frac{1}{2}$ **B.** $\frac{1}{6}$ **C.** $\frac{1}{3}$ **D.** $\frac{1}{4}$

Q3.04.1 MCQ 2004 1M

The distribution function $F_x(x)$ of a random variable X is shown in the figure. The probability that $X = 1$ is

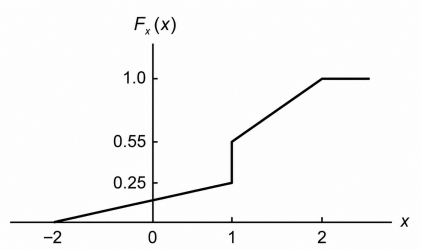


Fig. Q3.04.1

A. zero **B.** 0.25 **C.** 0.55 **D.** 0.30

Q3.03.1 MCQ 2003 2M

Let X and Y be two statistically independent random variables uniformly distributed in the ranges (-1,1) and (-2,1) respectively. Let $Z = X + Y$. Then the probability that $(z \leq -1)$ is

A. zero **B.** $\frac{1}{6}$ **C.** $\frac{1}{3}$ **D.** $\frac{1}{12}$

Q3.01.1 MCQ 2001 1M

The PDF of a Gaussian random variable X is given by $P_x(x) = \frac{1}{3\sqrt{2\pi}} e^{\frac{-(x-4)^2}{18}}$. The probability of the event $(X = 4)$ is

A. $\frac{1}{2}$ **B.** $\frac{1}{3\sqrt{2\pi}}$ **C.** 0 **D.** $\frac{1}{4}$

Q3.93.1 NAT 1993 2M

The function shown in figure, can represent a probability density function for A _____

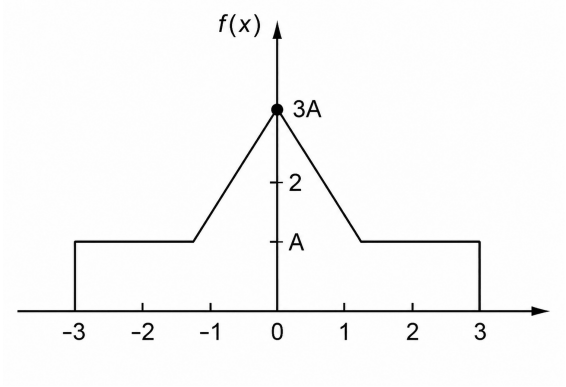


Fig. Q3.93.1

Q3.92.1 MCQ 1992 2M

For a random variable X following the probability density function $p(x)$, shown in given figure the mean and the variance are, respectively,

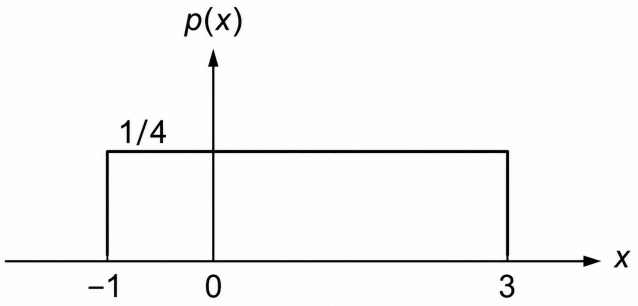


Fig. Q3.92.1

A. $\frac{1}{2}$ and $\frac{2}{3}$ **B.** 1 and $\frac{4}{3}$

C. 1 and $\frac{2}{3}$ **D.** 2 and $\frac{4}{3}$

Q3.88.1 MSQ 1988

Events A and B are mutually exclusive and have nonzero probability. Which of the following statement(s) are true?

- ☐ **A.** $P(A \cup B) = P(A) + P(B)$
- ☐ **B.** $P(B^C) > P(A)$
- ☐ **C.** $P(A \cap B) = P(A)P(B)$
- ☐ **D.** $P(B^C) < P(A)$

Electrical Engineering

Q3.26.1 NAT 2026 2M ○○○

Let X and Y be two real-valued random variables with $E(X) = 1$, $E(Y) = 2$, $E(X^2) = 4$, $E(Y^2) = 9$ and $E(XY) = 0.9$, where E denotes the expectation operator.The value of α that minimizes $E\left((X - \alpha Y)^2\right)$ is _____.

(Round off to one decimal place)

Q3.25.1 MCQ 2025 2M ○○○

Let X and Y be continuous random variables with probability density functions $P_X(x)$ and $P_Y(y)$, respectively. Further, let $Y = X^2$ and

$$P_X(x) = \begin{cases} 1, & x \in (0, 1] \\ 0, & \text{otherwise} \end{cases}$$

Which one of the following options is correct?

A. $P_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}}, & y \in (0, 1] \\ 0, & \text{otherwise} \end{cases}$

B. $P_Y(y) = \begin{cases} 1, & y \in (0, 1] \\ 0, & \text{otherwise} \end{cases}$

C. $P_Y(y) = \begin{cases} 1.5\sqrt{y}, & y \in (0, 1] \\ 0, & \text{otherwise} \end{cases}$

D. $P_Y(y) = \begin{cases} 2y, & y \in (0, 1] \\ 0, & \text{otherwise} \end{cases}$

Q3.25.2 MCQ 2025 1M ○○○

Consider discrete random variables X and Y with probabilities as follows:

$$P(X = 0 \text{ and } Y = 0) = \frac{1}{4}$$

$$P(X = 1 \text{ and } Y = 0) = \frac{1}{8}$$

$$P(X = 0 \text{ and } Y = 1) = \frac{1}{2}$$

$$P(X = 1 \text{ and } Y = 1) = \frac{1}{8}$$

Given $X = 1$, the expected value of Y is

A. $\frac{1}{4}$ **B.** $\frac{1}{2}$ **C.** $\frac{1}{8}$ **D.** $\frac{1}{3}$

Q3.24.1 MSQ 2024 2M ○○○

Let X be a discrete random variable that is uniformly distributed over the set $\{-10, -9, \ldots, 0, \ldots, 9, 10\}$.

Which of the following random variables is/are uniformly distributed?

A. X^2 **B.** X^3

C. $(X - 5)^2$ **D.** $(X + 10)^2$

Q3.23.1 NAT 2023 2M ○○○

The expected number of trials for first occurrence of a "head" in a biased coin is known to be 4. The probability of first occurrence of a "head" in the second trial is _____.

(Round off to 3 decimal places).

Q3.23.2 MCQ 2023 1M ○○○

One million random numbers are generated from a statistically stationary process with a Gaussian distribution with mean zero and standard deviation σ_0.

The σ_0 is estimated by randomly drawing out 10,000 numbers of samples (x_n). The estimates $\hat{\sigma}_1$, $\hat{\sigma}_2$ are computed in the following two ways.

$$\hat{\sigma}_1^2 = \frac{1}{10000} \sum_{n=1}^{10000} x_n^2$$

$$\hat{\sigma}_2^2 = \frac{1}{9999} \sum_{n=1}^{10000} x_n^2$$

Which of the following statements is true?

A. $E(\hat{\sigma}_2^2) = \sigma_0^2$ **B.** $E(\hat{\sigma}_2) = \sigma_0$

C. $E(\hat{\sigma}_1^2) = \sigma_0^2$ **D.** $E(\hat{\sigma}_1) = E(\hat{\sigma}_2)$

Q3.22.1 MCQ 2022 1M ○○○

Consider two dices having faces as P, Q, R, S, T, U, if both the dices are rolled together then what is the probability of getting any combination of P, Q, R on upper faces?

A. $\frac{1}{4}$ **B.** $\frac{1}{6}$ **C.** $\frac{5}{36}$ **D.** 0

Q3.22.2 NAT 2022 1M ○○○

Let the probability density function of a random variable x be given as

$$f(x) = ae^{-2|x|}$$

The value of a is _____.

Q3.21.1 MCQ 2021 2M

Suppose the probability that a coin toss shows "head" is p, where $0 < p < 1$. The coin is tossed repeatedly until the first "head" appears. The expected number of tosses required is

A. $\frac{p}{1-p}$ **B.** $\frac{1}{p}$

C. $\frac{1}{p^2}$ **D.** $\frac{1-p}{p}$

Q3.19.1 NAT 2019 2M

The probability of a resistor being defective is 0.02. There are 50 such resistors in a circuit. The probability of two or more defective resistors in the circuit (round off to two decimal places) is _____.

Q3.19.2 MCQ 2019 1M

The mean-square of a zero-mean random process is kT/C, where k is Boltzman's constant, T is the absolute temperature, and C is a capacitance. The standard deviation of the random process is

A. $\frac{\sqrt{kT}}{C}$ **B.** $\sqrt{\frac{kT}{C}}$

C. $\frac{C}{kT}$ **D.** $\frac{kT}{C}$

Q3.17.1 MCQ 2017 2M

A person decides to toss a fair coin repeatedly until he gets a head. He will make at most 3 tosses. Let the random variable Y denote the number of heads. The value of var$\{Y\}$, where var$[\cdot]$ denotes the variance, equals

A. $\frac{7}{8}$ **B.** $\frac{49}{64}$ **C.** $\frac{7}{64}$ **D.** $\frac{105}{64}$

Q3.17.2 MCQ 2017 1M

A urn contains 5 red ball and 5 black balls. In the first draw, one ball is picked at random and discarded without noticing its colour. The probability to get a red ball in the second draw is

A. $\frac{1}{2}$ **B.** $\frac{4}{9}$ **C.** $\frac{5}{9}$ **D.** $\frac{6}{9}$

Q3.17.3 NAT 2017 1M

Assume that in a traffic junction, the cycle of the traffic signal lights is 2 minutes of green (vehicle does not stop) and 3 minutes of red (vehicle stops). Consider that the arrival time of vehicles at the junction is uniformly distributed over 5 minute cycle. The expected waiting time (in minutes) for the vehicle at the junction is _____

Q3.16.1 MCQ 2016 2M

Let the probability density function of a variable, X, be given as

$$f_X(x) = \frac{3}{2}e^{-3x}u(x) + ae^{4x}u(-x)$$

Where $u(x)$ is the unit step function.
Then the value of 'a' and Prob$\{X \le 0\}$, respectively, are

A. 2, $\frac{1}{2}$ **B.** 4, $\frac{1}{2}$ **C.** 2, $\frac{1}{4}$ **D.** 4, $\frac{1}{4}$

Q3.16.2 NAT 2016 2M

Candidates were asked to come to an interview with 3 pens each. Black, blue, green and red were the permitted pen colours that the candidate could bring. The probability that a candidate comes with all 3 pens having the same colour is _____

Q3.15.1 MCQ 2015 2M

Two coins R and S are tossed. The 4 joint events H_RH_S, T_RT_S, H_RT_S, T_RH_S have probabilities 0.28, 0.18, 0.30, 0.24 respectively, where H represents head and T represents tail. Which one of the following is TRUE?

A. The coin tosses are independent.

B. R is fair, S is not.

C. S is fair, R is not.

D. The coin tosses are dependent.

Q3.15.2 MCQ 2015 2M

Two players, A and B, alternately keep rolling a fair dice. The person to get a six first wins the game. Given that player A starts the game, the probability that A wins the game is

A. $\frac{5}{11}$ **B.** $\frac{1}{2}$ **C.** $\frac{7}{13}$ **D.** $\frac{6}{11}$

Q3.15.3 NAT 2015 1M

A random variable X has probability density function $f(x)$ as given below:

$$f(x) = \begin{cases} a + bx & \text{for } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

If the expected value $E[X] = \frac{2}{3}$, then $P[X < 0.5]$ is _____

Q3.14.1 NAT 2014 1M

Lifetime of an electric bulb is a random variable with density $f(x) = kx^2$, where x is measured in years. If the minimum and maximum lifetimes of bulb are 1 and 2 years respectively, then the value of k is _____

Q3.14.2 NAT 2014 2M

Let X be a random variable with probability density function

$$f(x) = \begin{cases} 0.2, & \text{for } |x| \le 1 \\ 0.1, & \text{for } 1 < |x| \le 4 \\ 0, & \text{otherwise} \end{cases}$$

The probability $P(0.5 < X < 5)$ is _____

Q3.14.3 NAT 2014 1M

Consider a dice with the property that the probability of a face with n dots showing up is proportional to n. The probability of the face with three dots showing up is _____.

Q3.14.4 MCQ 2014 2M

A fair coin is tossed n times. The probability that the difference between the number of heads and tails is $(n-3)$ is

A. 2^{-n} **B.** 0

C. ${}^{n}C_{n-3}\,2^{-n}$ **D.** 2^{-n+3}

Q3.13.1 MCQ 2013 1M

A continuous random variable X has a probability density function

$$f(x) = e^{-x}, \quad 0 < x < \infty.$$

Then $P\{X > 1\}$ is

A. 0.368 **B.** 0.5 **C.** 0.632 **D.** 1.0

Q3.12.1 MCQ 2012 1M

A fair coin is tossed till a head appears for the first time. The probability that the number of required tosses is odd, is

A. $\frac{1}{3}$ **B.** $\frac{1}{2}$ **C.** $\frac{2}{3}$ **D.** $\frac{3}{4}$

Q3.12.2 MCQ 2012 1M

Two independent random variables X and Y are uniformly distributed in the interval $[-1, 1]$. The probability that $\max[X, Y]$ is less than $\frac{1}{2}$ is

A. $\frac{3}{4}$ **B.** $\frac{9}{16}$ **C.** $\frac{1}{4}$ **D.** $\frac{2}{3}$

Q3.11.1 MCQ 2011 2M

A zero mean random signal is uniformly distributed between limits $-a$ and $+a$ and its mean square value is equal to its variance. Then the r.m.s value of the signal is

A. $\frac{a}{\sqrt{3}}$ **B.** $\frac{a}{\sqrt{2}}$ **C.** $a\sqrt{2}$ **D.** $a\sqrt{3}$

Q3.10.1 MCQ 2010 2M

A box contains 4 white balls and 3 red balls. In succession, two balls are randomly selected and removed from the box. Given that the first removed ball is white, the probability that the second removed ball is red is

A. $\frac{1}{3}$ **B.** $\frac{3}{7}$ **C.** $\frac{1}{2}$ **D.** $\frac{4}{7}$

Q3.09.1 MCQ 2009 2M

Assume for simplicity that N people, all born in April (a month of 30 days) are collected in a room, consider the event of at least two people in the room being born on the same date of the month even if in different years e.g. 1980 and 1985. What is the smallest N so that the probability of this exceeds 0.5 is?

A. 20 **B.** 7 **C.** 15 **D.** 16

Q3.08.1 MCQ 2008 1M

X is uniformly distributed random variable that take values between 0 and 1. The value of $E(X^3)$ will be,

A. 0 **B.** $\frac{1}{8}$ **C.** $\frac{1}{4}$ **D.** $\frac{1}{2}$

Q3.07.1 MCQ 2007 1M

A loaded dice has following probability distribution of occurrences

Dice value	1	2	3	4	5	6
Probability	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$

If three identical dice as the above are thrown, the probability of occurrence of values 1, 5 and 6 on the three dice is

A. Same as that of occurrence of 3, 4, 5.

B. Same as that of occurrence of 1, 2, 5.

C. $\frac{1}{128}$

D. $\frac{5}{8}$

Q3.06.1 MCQ 2006 2M

Two fair dice are rolled and the sum r of the numbers turned up is considered

A. $Pr(r > 6) = \left(\frac{1}{6}\right)$

B. $Pr(r/3 \text{ is an integer}) = \left(\frac{5}{6}\right)$

C. $Pr(r = 8 \mid r/4 \text{ is an integer}) = \left(\frac{5}{9}\right)$

D. $Pr(r = 6 \mid r/5 \text{ is an integer}) = \left(\frac{1}{18}\right)$

Q3.05.1 MCQ 2005 1M

If P and Q are two random events, then which of the following is true?

A. Independence of P and Q implies that Probability $(P \cap Q) = 0$

B. Probability $(P \cap Q) \geq$ Probability (P)+Probability (Q)

C. If P and Q are mutually exclusive then they must be independent

D. Probability $(P \cap Q) \leq$ Probability (P)

Q3.05.2 MCQ 2005 2M

A fair coin is tossed three times in succession. If the first toss produces a head, then the probability of getting exactly two heads in three tosses is

A. $\frac{1}{8}$ **B.** $\frac{1}{2}$ **C.** $\frac{3}{8}$ **D.** $\frac{3}{4}$

Instrumentation Engineering

Q3.26.1 NAT 2026 1M

A data set is given to be $[1, 2, 0, -1, -3, 1, 2, 0, 1]$. The median of the data set is_____. (rounded off to the nearest integer)

Q3.26.2 MCQ 2026 2M

A box contains three apples and two oranges. Two fruits are removed randomly in succession. The probability that the first is an apple and the second is an orange is_____.

A. $\frac{3}{5}$ **B.** $\frac{3}{20}$ **C.** $\frac{3}{10}$ **D.** $\frac{2}{5}$

Q3.25.1 NAT 2025 2M

The probability of a student missing a class is 0.1. In a total number of 10 classes, the probability that the student will not miss more than one class is _____. (rounded off to two decimal places).

Q3.24.1 NAT 2024 1M

Indian Premier League has divided the sixteen cricket teams into two equal pools : Pool-A and Pool-B. Four teams of Pool-A have blue logo jerseys while the rest four have red logo jerseys. Five teams of Pool-B have blue logo jerseys while the rest three have red logo jerseys. If one team from each pool reaches the final, the probability that one team has a blue logo jersey and another has a red logo jersey is _____ (rounded off to one decimal place).

Q3.24.2 NAT 2024 2M

A random variable X has a probability density function

$$f_X(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

The probability of $X > 2$ is _____ (rounded off to three decimal places).

Q3.23.1 NAT 2023 1M

X is a discrete random variable which takes values 0, 1 and 2. The probabilities are $P(X = 0) = 0.25$ and $P(X = 1) = 0.5$. With $E[\cdot]$ denoting the expectation operator, the value of $E[X] - E[X^2]$ is _____ (rounded off to one decimal places).

Q3.23.2 NAT 2023 2M

How many five digit numbers can be formed using the integers 3, 4, 5 and 6 with exactly one digit appearing twice?

Q3.22.1 NAT 2022 1M

A box contains 6 red and 4 blue balls. If three ball are drawn in succession without replacement. Then, the probability of 2nd and 3rd ball of having color red is _____.

Q3.20.1 NAT 2020 1M

A player throws a ball at a basket kept at a distance. The probability that the ball falls into the basket in a single attempt is 0.1. The player attempts to throw the ball twice. Considering each attempt to be independent, the probability that this player puts the ball into the basket only in the second attempt (rounded off to two decimal places) is _____.

Q3.20.2 NAT 2020 2M

Consider two identical bags B_1 and B_2, each containing 10 balls of identical shapes and sizes. Bag B_1 contains 7 Red and 3 Green balls, while bag B_2 contains 3 Red and 7 Green balls. A bag is picked at random and a ball is drawn from it, which was found to be Red. The probability that the Red ball came from bag B_1 (rounded off to one decimal place) is _____.

Q3.19.1 MCQ 2019 2M

The function $p(x)$ is given by

$$p(x) = \frac{A}{x^{\mu}}$$

where A and μ are constants with $\mu > 1$ and $1 \leq x < \infty$ and $p(x) = 0$ for $-\infty < x < 1$. For $p(x)$ to be probability density function, the value of A should be equal to

A. $\mu - 1$ **B.** $\mu + 1$

C. $\frac{1}{(\mu - 1)}$ **D.** $\frac{1}{(\mu + 1)}$

Q3.19.2 MCQ 2019 1M

A box has 8 red balls and 8 green balls. Two balls are drawn randomly in succession from the box without replacement. The probability that the first ball drawn is red and the second ball drawn is green is

A. $\frac{4}{15}$ **B.** $\frac{7}{16}$ **C.** $\frac{1}{2}$ **D.** $\frac{8}{15}$

Q3.18.1 MCQ 2018 1M

X and Y are two independent random variables with variances 1 and 2, respectively. Let $Z = X - Y$. The variance of Z is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q3.18.2 MCQ 2018 1M

Consider a sequence of tossing of a fair coin where the outcomes of tosses are independent. The probability of getting the head for the third time in the fifth toss is

A. $\frac{5}{16}$ **B.** $\frac{3}{16}$ **C.** $\frac{3}{5}$ **D.** $\frac{9}{16}$

Q3.18.3 MCQ 2018 2M

Two bags A and B have equal number of balls. Bag A has 20% red balls and 80% green balls. Bag B has 30% red balls, 60% green balls and 10% yellow balls. Contents of Bags A and B are mixed thoroughly and a ball is randomly picked from the mixture. What is the chance that the ball picked is red?

A. 20% **B.** 25% **C.** 30% **D.** 40%

Q3.17.1 MCQ 2017 2M

The probability that a communication system will have high fidelity is 0.81. The probability that the system will have both high fidelity and high selectivity is 0.18. The probability that a given system with high fidelity will have high selectivity is

A. 0.181 **B.** 0.191 **C.** 0.222 **D.** 0.826

Q3.16.1 MCQ 2016 2M

An urn contains 5 red and 7 green balls. A ball is drawn at random and its colour is noted. The ball is placed back into the urn along with another ball of the same colour. The probability of getting a red ball in the next draw is

A. $\frac{65}{156}$ **B.** $\frac{67}{156}$ **C.** $\frac{79}{156}$ **D.** $\frac{89}{156}$

Q3.15.1 MCQ 2015 2M

The probability that a thermistor randomly picked up from a production unit is defective is 0.1. The probability that out of 10 thermistors, randomly picked up 3 are defective is

A. 0.001 **B.** 0.057 **C.** 0.107 **D.** 0.3

Q3.15.2 NAT 2015 2M

The probability density function of a random variable X is $P_X(x) = e^{-x}$ for $x \geq 0$ and 0 otherwise. The expected value of the function $g_2(x) = e^{3x/4}$ is _____.

Q3.14.1 NAT 2014 1M

The figure shows the schematic of a production process with machines A, B and C. An input job needs to be pre-processed either by A or by B before it is fed to C, from which the final finished product comes out. The probabilities of failure of the machines are given as

$$P_A = 0.15, \qquad P_B = 0.05, \qquad P_C = 0.1.$$

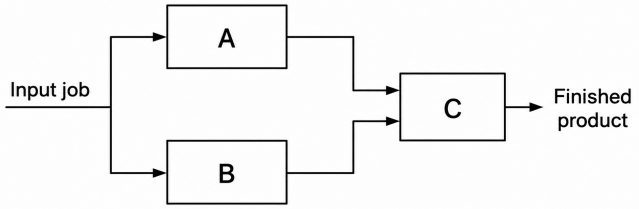


Fig. Q3.14.1

Assuming independence of failures of the machines, the probability that a given job is successfully processed (up to the third decimal place) is _____.

Q3.14.2 NAT 2014 1M

Given that x is a random variable in the range $[0, \infty]$ with a probability density function

$$\frac{e^{-x/2}}{K},$$

the value of the constant K is _____.

Q3.13.1 MCQ 2013 1M

A continuous random variable X has a probability density function

$$f(x) = e^{-x}, \ 0 < x < \infty$$

Then $P(X > 1)$ is

A. 0.368 **B.** 0.5 **C.** 0.632 **D.** 1.0

Q3.12.1 MCQ 2012 1M

Two independent random variables X and Y are uniformly distributed in the interval $[-1, 1]$. The probability that $\max[X, Y]$ is less than $\frac{1}{2}$ is

A. $\frac{3}{4}$ **B.** $\frac{9}{16}$ **C.** $\frac{1}{4}$ **D.** $\frac{2}{3}$

Q3.11.1 MCQ 2011 2M

The box 1 contains chips numbered 3, 6, 9, 12 and 15. The box 2 contains chips numbered 6, 11, 16, 21 and 26. Two chips, one from each box, are drawn at random. The numbers written on these chips are multiplied. The probability for the product to be an even number is

A. $\frac{6}{25}$ **B.** $\frac{2}{5}$ **C.** $\frac{3}{5}$ **D.** $\frac{19}{25}$

Q3.10.1 MCQ 2010 1M

The diameters of 10000 ball bearings were measured. The mean diameter and standard deviation were found to be 10 mm and 0.05 mm respectively. Assuming Gaussian distribution of measurements, it can be expected that the number of measurements more than 10.15 mm will be

A. 230 **B.** 115 **C.** 15 **D.** 2

Q3.09.1 MCQ 2009 2M

A screening test is carried out to detect a certain disease. It is found that 12% of the positive reports and 15% of the negative reports are incorrect. Assuming that the probability of a person getting a positive report is 0.01, the probability that a person tested gets incorrect report is

A. 0.0027 **B.** 0.0173 **C.** 0.1497 **D.** 0.2100

Q3.08.1 MCQ 2008 2M

A random variable is uniformly distributed over the interval 2 to 10. Its variance will be

A. $\frac{16}{3}$ **B.** 6 **C.** $\frac{256}{9}$ **D.** 36

Q3.08.2 MCQ 2008 2M

Consider a Gaussian distributed random variable with zero mean and standard deviation σ. The value of its cumulative distribution function at the origin will be

A. 0 **B.** 0.5 **C.** 1 **D.** 10

Q3.07.1 MCQ 2007 1M

Assume that the duration in minutes of a telephone conversation follows the exponential distribution

$$f(x) = \frac{1}{5}e^{-x/5}, \qquad x \geq 0.$$

The probability that the conversation will exceed five minutes is

A. $\frac{1}{e}$ **B.** $1 - \frac{1}{e}$ **C.** $\frac{1}{e^2}$ **D.** $1 - \frac{1}{e^2}$

Q3.06.1 MCQ 2006 2M

You have gone to a cyber-cafe with a friend. You found that the cyber-cafe has only three terminals. All terminals are unoccupied. You and your friend have to make a random choice of selecting a terminal. What is the probability that both of you will NOT select the same terminal?

A. $\frac{1}{9}$ **B.** $\frac{1}{3}$ **C.** $\frac{2}{3}$ **D.** 1

Q3.06.2 MCQ 2006 2M

Probability density function $p(x)$ of a random variable x is as shown below. The value of α is

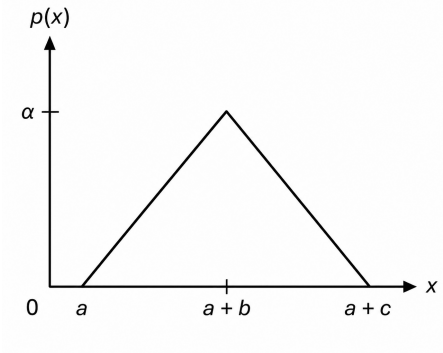


Fig. Q3.06.2

A. $\frac{2}{c}$ **B.** $\frac{1}{c}$

C. $\frac{2}{(b+c)}$ **D.** $\frac{1}{(b+c)}$

Q3.06.3 MCQ 2006 1M

Two dices are rolled simultaneously. The probability that the sum of digits on the top surface of the two dices is even, is

A. 0.5 **B.** 0.25 **C.** 0.167 **D.** 0.125

Q3.05.1 MCQ 2005 1M

The probability that there are 53 Sundays in a randomly chosen leap year is

A. $\frac{1}{7}$ **B.** $\frac{1}{14}$ **C.** $\frac{1}{28}$ **D.** $\frac{2}{7}$

Computer Science & IT

Q3.26.1 MCQ 2026 2M

An unbiased six-faced dice whose faces are marked with numbers $1, 2, 3, 4, 5,$ and 6 is rolled twice in succession and the number on the top face is recorded each time. The probability that the number appearing in the second roll is an integer multiple of the number appearing in the first roll is _____

A. $\frac{1}{6}$ **B.** $\frac{5}{18}$ **C.** $\frac{7}{18}$ **D.** $\frac{5}{6}$

Q3.26.2 MCQ 2026 1M

An urn contains one red ball and one blue ball. At each step, a ball is picked uniformly at random from the urn, and this ball together with another ball of the same color is put back in the urn. The probability that there are equal number of red and blue balls after two steps is

A. $\frac{1}{4}$ **B.** $\frac{1}{3}$ **C.** $\frac{1}{2}$ **D.** $\frac{2}{3}$

Q3.26.3 NAT 2026 2M

Let X be a random variable which takes values in the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$. Further,

$$\Pr(X = 1) = \Pr(X = 2) = \Pr(X = 5) = \Pr(X = 7) = \frac{1}{6}$$

and

$$\Pr(X = 3) = \Pr(X = 4) = \Pr(X = 6) = \Pr(X = 8) = \frac{1}{12}.$$

The expected value of X, denoted by $E[X]$, is equal to _____. (rounded off to two decimal places)

Q3.26.4 MCQ 2026 1M

A day can only be cloudy or sunny. The probability of a day being cloudy is 0.5, independent of the condition on other days. What is the probability that in any given four days, there will be three cloudy days and one sunny day?

A. $\frac{1}{4}$ **B.** $\frac{3}{4}$ **C.** $\frac{2}{3}$ **D.** $\frac{3}{8}$

Q3.26.5 MCQ 2026 1M

The probability density function $f(x)$ of a random variable X which takes real values is

$$f(x) = \frac{1}{3\sqrt{2\pi}} \exp\left(-\frac{x^2}{18}\right), \qquad x \in (-\infty, +\infty)$$

Which one of the following statements is correct about the random variable X?

A. X is an exponential random variable

B. X is a normal random variable

C. X is a Poisson random variable

D. X is a uniform random variable

Q3.26.6 NAT 2026 2M

Suppose an unbiased coin is tossed 6 times. Each coin toss is independent of all previous coin tosses. Let E_1 be the event that among the second, fourth, and sixth coin tosses, there are at least two heads. Let E_2 be the event that among the first, second, third, and fifth coin tosses, there are equal number of heads and tails.
The conditional probability $P(E_1 \mid E_2)$ is equal to _____ (rounded off to one decimal place)

Q3.25.1 NAT 2025 2M

A quadratic polynomial $(x-\alpha)(x-\beta)$ over complex numbers is said to be square invariant if $(x-\alpha)(x-\beta) = (x-\alpha^2)(x-\beta^2)$.. Suppose from the set of all square invariant quadratic polynomials we choose one at random.The probability that the roots of the chosen polynomial are equal is _____(rounded off to one decimal place)

Q3.25.2 NAT 2025 2M

The unit interval $(0, 1)$ is divided at a point chosen uniformly distributed over $(0, 1)$ in $\mathbb{R}$ into two disjoint subintervals.
The expected length of the subinterval that contains 0.4 is _____. (rounded off to two decimal places)

Q3.25.3 NAT 2025 1M

A box contains 5 coins: 4 regular coins and 1 fake coin. When a regular coin is tossed, the probability $P(\text{head}) = 0.5$ and for a fake coin, $P(\text{head}) = 1$. You pick a coin at random and toss it twice, and get two heads. The probability that the coin you have chosen is the fake coin is _____. (Rounded off to two decimal places)

Q3.25.4 NAT 2025 2M

Consider a probability distribution given by the density function $P(x)$.

$$P(x) = \begin{cases} Cx^2, & \text{for } 1 \le x \le 4 \\ 0, & \text{for } x < 1 \text{ or } x > 4 \end{cases}$$

The probability that x lies between 2 and 3, i.e., $P(2 \le x \le 3)$ is _____. (Rounded off to three decimal places)

Q3.24.1 MCQ 2024 1M

When six unbiased dice are rolled simultaneously, the probability of getting all distinct numbers (i.e. $1, 2, 3, 4, 5$ and 6) is

A. $\frac{5}{324}$ **B.** $\frac{11}{324}$ **C.** $\frac{1}{324}$ **D.** $\frac{7}{324}$

Q3.24.2 MCQ 2024 2M

Let x and y be random variables, not necessarily independent, that take real values in the interval $[0, 1]$. Let $z = xy$ and let the mean values of x, y, z be $\bar{x}, \bar{y}, \bar{z}$ respectively. Which one of the following statements is TRUE?

A. $\bar{z} = \bar{x}\bar{y}$ **B.** $\bar{z} \ge \bar{x}\bar{y}$

C. $\bar{z} \le \bar{x}\bar{y}$ **D.** $\bar{z} \le \bar{x}$

Q3.24.3 NAT 2024 2M

A bag contains 10 red balls and 15 blue balls. Two balls are drawn randomly without replacement. Given that the first ball drawn is red, the probability (rounded off to 3 decimal places) that both balls drawn are red is _____.

Q3.24.4 MSQ 2024 1M

Let A and B be two events in a probability space with $P(A) = 0.3$, $P(B) = 0.5$, and $P(A \cap B) = 0.1$. Which of the following statement is/are TRUE?

☐ **A.** $P(A \cap B^c) = 0.2$, where B^c is the complement of the event B

☐ **B.** The two events A and B are independent

☐ **C.** $P(A \cup B) = 0.7$

☐ **D.** $P(A^c \cap B^c) = 0.4$, where A^c and B^c are the complements of the events A and B, respectively

Q3.24.5 MCQ 2024 1M

Consider a permutation sampled uniformly at random from the set of all permutations of $1, 2, 3, \ldots, n$ for some $n \geq 4$. Let X be the event that 1 occurs before 2 in the permutation, and Y the event that 3 occurs before 4. Which one of the following statements is TRUE?

A. Event X is more likely than event Y

B. The events X and Y are independent

C. Either event X or Y must occur

D. The events X and Y are mutually exclusive

Q3.23.1 MSQ 2023 2M

Consider a random experiment where two fair coins are tossed. Let A be the event that denotes HEAD on both the throws, B be the event that denotes HEAD on the first throw, and C be the event that denotes HEAD on the second throw. Which of the following statements is/are TRUE?

☐ **A.** A and B are independent.

☐ **B.** A and C are independent.

☐ **C.** B and C are independent.

☐ **D.** $\text{Prob}(B \mid C) = \text{Prob}(B)$

Q3.21.1 MCQ 2021 1M

The shape of the cumulative distribution function of Gaussian distribution is

A. Bell-shaped **B.** S-shaped

C. Horizontal line **D.** Straight line at 45 degree angle

Q3.21.2 MCQ 2021 2M

A bag has r red balls and b black balls. All balls are identical except for their colours. In a trial, a ball is randomly drawn from the bag, its colour is noted and the ball is placed back into the bag along with another ball of the same colour. Note that the number of balls in the bag will increase by one, after the trial. A sequence of four such trials is conducted. Which one of the following choices gives the probability of drawing a red ball in the fourth trial?

A. $\dfrac{r}{r+b}$

B. $\dfrac{r}{r+b+3}$

C. $\left(\dfrac{r}{r+b}\right)\left(\dfrac{r+1}{r+b+1}\right)\left(\dfrac{r+2}{r+b+2}\right)\left(\dfrac{r+3}{r+b+3}\right)$

D. $\dfrac{r+3}{r+b+3}$

Q3.21.3 MCQ 2021 2M

In an examination, a student can choose the order in which two questions (QuesA and QuesB) must be attempted.

- If the first question is answered wrong, the student gets zero marks.
- If the first questions is answered correctly and the second question is not answered correctly, the student gets the marks only for the first question.
- If both the questions are answered correctly, the student gets the sum of the marks of the two questions.

The following table shows the probability of correctly answering a question and the marks of the question respectively.

Question	Probability of answering correctly	Marks
QuesA	0.8	10
QuesB	0.5	20

Assuming that the student always wants to maximize her expected marks in the examination, in which order should she attempt the questions and what is the expected marks for that order (assume that the questions are independent)?

A. First QuesA and then QuesB. Expected marks 14.

B. First QuesB and then QuesA. Expected marks 22.

C. First QuesB and then QuesA. Expected marks 14.

D. First QuesA and then QuesB. Expected marks 16.

Q3.21.4 NAT 2021 1M

For a given biased coin, the probability that the outcome of a toss is a head is 0.4. This coin is tossed 1,000 times. Let X denote the random variable whose value is the number of times that head appeared in these 1,000 tosses. The standard deviation of X (rounded to 2 decimal places) is _____.

Q3.21.5 NAT 2021 2M

A sender (S) transmits a signal, which can be one of the two kinds: H and L with probability 0.1 and 0.9 respectively, to a receiver (R). In the graph below, the weight of edge (u, v) is the probability of receiving v when u is transmitted, where $u, v \in \{H, L\}$. For example, the probability that the

received signal is L given the transmitted signal was H, is 0.7.

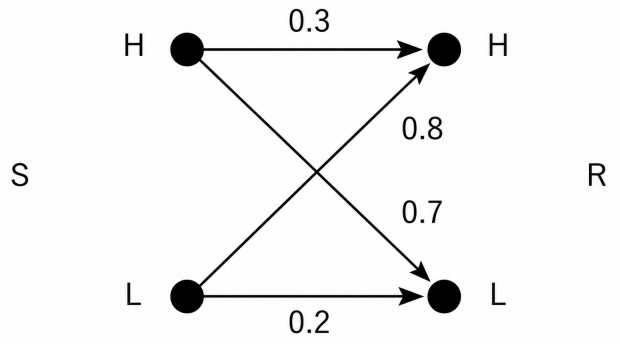


Fig. Q3.21.5

If the received signal is H, the probability that the transmitted signal was H (rounded to 2 decimal places) is _____.

Q3.20.1 NAT 2020 2M

For $n > 2$, let $a \in \{0,1\}^n$ be a non-zero vector. Suppose that x is chosen uniformly at random from $\{0,1\}^n$. Then, the probability that $\sum_{i=1}^{n} a_i x_i$ is an odd number is _____.

Q3.19.1 NAT 2019 1M

Two numbers are chosen independently and uniformly at random from the set $\{1, 2, \ldots, 13\}$. The probability (rounded off to 3 decimal places) that their 4-bit (unsigned) binary representations have the same most significant bit is _____.

Q3.19.2 NAT 2019 2M

Suppose Y is distributed uniformly in the open interval $(1, 6)$. The probability that the polynomial $3x^2 + 6xY + 3Y + 6$ has only real roots is (rounded off to 1 decimal place) _____.

Q3.18.1 NAT 2018 1M

Two people, P and Q, decide to independently roll two identical dice, each with 6 faces, numbered 1 to 6. The person with the lower number wins. In case of a tie, they roll the dice repeatedly until there is no tie. Define a trial as a throw of the dice by P and Q. Assume that all 6 numbers on each dice are equiprobable and that all trials are independent. The probability (rounded to 3 decimal places) that one of them wins on the third trial is _____.

Q3.18.2 NAT 2018 2M

Consider Guwahati (G) and Delhi (D) whose temperatures can be classified as high (H), medium (M) and low (L). Let $P(H_G)$ denote the probability that Guwahati has high temperature. Similarly, $P(M_G)$ and $P(L_G)$ denotes the probability of Guwahati having medium and low temperatures respectively. Similarly, we use $P(H_D)$, $P(M_D)$ and $P(L_D)$ for Delhi.

The following table gives the conditional probabilities for Delhi's temperature given Guwahati's temperature.

	H_D	M_D	L_D
H_G	0.40	0.48	0.12
M_G	0.10	0.65	0.25
L_G	0.01	0.50	0.49

Consider the first row in the table above. The first entry denotes that if Guwahati has high temperature (H_G) then the probability of Delhi also having a high temperature (H_D) is 0.40; i.e., $P(H_D \mid H_G) = 0.40$. Similarly, the next two entries are $P(M_D \mid H_G) = 0.48$ and $P(L_D \mid H_G) = 0.12$. Similarly for the other rows.

If it is known that $P(H_G) = 0.2$, $P(M_G) = 0.5$ and $P(L_G) = 0.3$, then the probability (correct to two decimal places) that Guwahati has high temperature given that Delhi has high temperature is _____.

Q3.17.1 NAT 2017 2M

If a random variable X has a Poisson distribution with mean 5, then the expectation $E[(X+2)^2]$ equals _____.

Q3.17.2 NAT 2017 1M

Let X be a Gaussian random variable with mean 0 and variance σ^2. Let $Y = \max(X, 0)$ where $\max(a, b)$ is the maximum of a and b. The median of Y is _____.

Q3.17.3 MCQ 2017 1M

The probability that a k-digit number does NOT contain the digits 0, 5, or 9 is

A. 0.3^k **B.** 0.6^k **C.** 0.7^k **D.** 0.9^k

Q3.16.1 NAT 2016 1M

A probability density function on the interval $[a, 1]$ is given by $\frac{1}{x^2}$ and outside this interval the value of the function is zero. The value of a is ____.

Q3.16.2 NAT 2016 2M

Consider the following experiment.
Step 1. Flip a fair coin twice.
Step 2. If the outcomes are (TAILS, HEADS) then output Y and stop.
Step 3. If the outcomes are either (HEADS, HEADS) or (HEADS, TAILS), then output N and stop.
Step 4. If the outcomes are (TAILS, TAILS), then go to Step 1.
The probability that the output of the experiment is Y is (up to two decimal places) _____ .

Q3.16.3 NAT 2016 1M

Suppose that a shop has an equal number of LED bulbs of two different types. The probability of an LED bulb lasting more than 100 hours given that it is of Type 1 is 0.7, and given that it is of Type 2 is 0.4. The probability that an LED bulb chosen uniformly at random lasts more than 100 hours is _____ .

Q3.14.1 NAT 2014 1M

The security system at an IT office is composed of 10 computers of which exactly four are working. To check whether the system is functional, the officials inspect four of the computers picked at random (without replacement). The system is deemed functional if at least three of the four computers inspected are working. Let the probability that the system is deemed functional be denoted by p. Then $100p$ = _____.

Q3.14.2 NAT 2014 2M

Let S be a sample space and two mutually exclusive events A and B be such that $A \cup B = S$. If $P(\cdot)$ denotes the probability of the event, the maximum value of $P(A)\,P(B)$ is _____.

Q3.14.3 NAT 2014 2M

The probability that a given positive integer lying between 1 and 100 (both inclusive) is NOT divisible by 2, 3 or 5 is ____.

Q3.14.4 NAT 2014 2M

Four fair six-sided dice are rolled. The probability that the sum of the results being 22 is $X/1296$.
The value of X is ____.

Q3.14.5 NAT 2014 1M

Each of the nine words in the sentence "The quick brown fox jumps over the lazy dog" is written on a separate piece of paper. These nine pieces of paper are kept in a box. One of the pieces is drawn at random from the box. The expected length of the word drawn is _____. (The answer should be rounded to one decimal place.)

Q3.14.6 NAT 2014 1M

Suppose you break a stick of unit length at a point chosen uniformly at random. Then the expected length of the shorter stick is _____.

Q3.13.1 MCQ 2013 1M

Suppose p is the number of cars per minute passing through a certain road junction between 5 PM, and p has Poisson distribution with mean 3. What is the probability of observing fewer than 3 cars during any given minute in this interval?

A. $8/(2e^3)$ **B.** $9/(2e^3)$
C. $17/(2e^3)$ **D.** $26/(2e^3)$

Q3.12.1 MCQ 2012 1M

Consider a random variable X that takes values $+1$ and -1 with probability 0.5 each. The values of the cumulative distribution function $F(x)$ at $x = -1$ and $+1$ are

A. 0 and 0.5 **B.** 0 and 1
C. 0.5 and 1 **D.** 0.25 and 0.75

Q3.12.2 MCQ 2012 2M

Suppose a fair six-sided die is rolled once. If the value on the die is 1, 2 or 3 the die is rolled a second time. What is the probability that the sum of total values that turn up is at least 6?

A. 10/21 **B.** 5/12 **C.** 2/3 **D.** 1/6

Q3.11.1 MCQ 2011 1M

If the difference between the expectation of the square of a random variable ($E[x^2]$) and the square of the expectation of the random variable $(E[x])^2$ is denoted by R, then

A. $R = 0$ **B.** $R < 0$ **C.** $R \geq 0$ **D.** $R > 0$

Q3.11.2 MCQ 2011 2M

Consider the finite sequence of random values $X = [x_1, x_2, \ldots, x_n]$. Let μ_x be the mean and σ_x be the standard deviation of X. Let another finite sequence Y of equal length be derived from this as $y_i = a * x_i + b$, where a and b are positive constant. Let μ_y be the mean and σ_y be the standard deviation of this sequence. Which one of the following statements INCORRECT?

A. Index position of mode of X in X is the same as the index position of mode of Y in Y.

B. Index position of median of X in X is the same as the index position of median of Y in Y.

C. $\mu_y = a\mu_x + b$

D. $\sigma_y = a\sigma_x + b$

Q3.11.3 MCQ 2011 2M

A deck of 5 cards (each carrying a distinct number from 1 to 5) is shuffled thoroughly. Two cards are then removed one at a time from the deck. What is the probability that the two cards are selected with the number on the first card being one higher than the number on the second card?

A. 1/5 **B.** 4/25 **C.** 1.4 **D.** 2/5

Q3.10.1 MCQ 2010 2M

Consider a company that assembles computers. The probability of a faulty assembly of any computer is p. The company therefore subjects each computer to a testing process. This testing process gives the correct result for any computer with a probability of q. What is the probability of a computer being declared faulty?

A. $pq + (1-p)(1-q)$ **B.** $(1-q)p$

C. $(1-p)q$ **D.** pq

Q3.09.1 MCQ 2009 2M

An unbalanced dice (with 6 faces, numbered from 1 to 6) is thrown. The probability that the face value is odd is 90% of the probability that the face value is even. The probability of getting any even numbered face is the same.

If the probability that the face is even given that it is greater than 3 is 0.75, which one of the following options is closest to the probability that the face value exceeds 3?

A. 0.453 **B.** 0.468 **C.** 0.485 **D.** 0.492

Q3.09.2 MCQ 2009 1M

If two fair coins are flipped and at least one of the outcomes is known to be a head, what is the probability that both outcomes are heads?

A. 1/3 **B.** 1/4 **C.** 1/2 **D.** 2/3

Q3.08.1 MCQ 2008 2M

Let X be a random variable following normal distribution with mean +1 and variance 4. Let Y be another normal variable with mean −1 and variance unknown,. If $P(X \leq -1) = P(Y \geq 2)$ the standard deviation of Y is

A. 3 **B.** 2 **C.** $\sqrt{2}$ **D.** 1

Q3.08.2 MCQ 2008 2M

Aishwarya studies either computer science or mathematics everyday. if she studies computer science on a day, then the probability that she studies mathematics the next day is 0.6. If she studies mathematics on a day, then the probability that she studies computer science the next day is 0.4. Given that Aishwarya studies computer science on Monday, what is the probability that she studies computer science on Wednesday?

A. 0.24 **B.** 0.36 **C.** 0.4 **D.** 0.6

Q3.07.1 MCQ 2007 2M

Suppose we uniformly and randomly select a permutation from the 20! permutations of $1, 2, 3, \ldots, 20$. What is the probability that 2 appears at an earlier position that any other even number in the selected permutation?

A. $\frac{1}{2}$ **B.** $\frac{1}{10}$

C. $\frac{9!}{20!}$ **D.** None of these

Q3.06.1 MCQ 2006 2M

For each element in a set of size $2n$, an unbiased coin is tossed. All the $2n$ coin tossed are independent. An element is chosen if the corresponding coin toss were head. The probability that exactly n elements are chosen is

A. $\binom{2n}{n}/4^n$ **B.** $\binom{2n}{n}/2^n$ **C.** $1/\binom{2n}{n}$ **D.** $\frac{1}{2}$

Q3.05.1 MCQ 2005 1M

Let $f(x)$ be the continuous probability density function of a random variable X. The probability that $a < X \leq b$, is

A. $f(b-a)$ **B.** $f(b) - f(a)$

C. $\int_a^b f(x)\,dx$ **D.** $\int_a^b xf(x)\,dx$

Q3.04.1 MCQ 2004 2M

A point is randomly selected with uniform probability in the X-Y plane within the rectangle with corners at $(0,0)$, $(1,0)$, $(1,2)$ and $(0,2)$. If p is the length of the position vector of the point, the expected value of p^2 is

A. $2/3$ **B.** 1 **C.** $4/3$ **D.** $5/3$

Q3.04.2 MCQ 2004 2M

An examination paper has 150 multiple-choice questions of one mark each, with each question having four choices. Each incorrect answer fetches -0.25 mark. Suppose 1000 students choose all their answers randomly with uniform probability. The sum total of the expected marks obtained by all these students is

A. 0 **B.** 2550 **C.** 7525 **D.** 9375

Q3.04.3 MCQ 2004 1M

If a fair coin is tossed four times. What is the probability that two heads and two tails will result?

A. $3/8$ **B.** $1/2$ **C.** $5/8$ **D.** $3/4$

Q3.04.4 MCQ 2004 2M

Two n bit binary strings, S_1 and S_2 are chosen randomly with uniform probability. The probability that the Hamming distance between these strings (the number of bit positions where the two strings differ) is equal to d is

A. ${}^nC_d/2^n$ **B.** ${}^nC_d/2^d$

C. $d/2^n$ **D.** $1/2^d$

Q3.03.1 MCQ 2003 2M

A program consists of two modules executed sequentially. Let $f_1(t)$ and $f_2(t)$ respectively denote the probability density functions of time taken to execute the two modules. The probability density function of the overall time taken to execute the program is given by

A. $f_1(t) + f_2(t)$ **B.** $\int_0^t f_1(x) f_2(x)\,dx$

C. $\int_0^t f_1(x) f_2(t-x)\,dx$ **D.** $\max\{f_1(t),\ f_2(t)\}$

Q3.03.2 MCQ 2003 1M

Let $P(E)$ denote the probability of the event E. Given $P(A) = 1$, $P(B) = 1/2$, the values of $P(A/B)$ and $P(B/A)$ respectively are

A. $1/4,\ 1/2$ **B.** $1/2,\ 1/4$

C. $1/2,\ 1$ **D.** $1,\ 1/2$

Mechanical Engineering

Q3.26.1 MCQ 2026 2M

A student council of 10 members consists of two students from engineering school, three students from science school and five students from arts school. The university administration selects three students from the council at random. What is the chance that out of the selected students, two belong to the same school and the third belongs to different school?

A. $\frac{79}{120}$ **B.** $\frac{2}{120}$ **C.** $\frac{89}{120}$ **D.** $\frac{69}{120}$

Q3.25.1 MCQ 2025 1M

If two unbiased coins are tossed, then what is the probability of having at least one head?

A. 0.25 **B.** 0.5 **C.** 0.675 **D.** 0.75

Q3.24.1 NAT 2024 2M

Let X be a continuous random variable defined on $[0, 1]$ such that its probability density function $f(x) = 1$ for $0 \leq x \leq 1$ and 0 otherwise. Let $Y = \log_e(X + 1)$.
Then the expected value of Y is _____ (rounded off to 2 decimal places).

Q3.23.1 MCQ 2023 1M

A machine produces a defective component with a probability of 0.015. The number of defective components in a packed box containing 200 components produced by the machine follows a Poisson distribution. The mean and the variance of the distribution are

A. 3 and 3, respectively

B. $\sqrt{3}$ and $\sqrt{3}$ respectively

C. 0.015 and 0.015, respectively

D. 3 and 9, respectively

Q3.22.1 NAT 2022 2M

Let a random variable x follows poisson's distribution such that prob$(x = 1)$ = prob$(x = 2)$. The value of prob$(x = 3)$ is _____. (Round off to 2 decimal places)

Q3.21.1 MCQ 2021 1M

The Mean and variance, respectively, of a binomial distribution for n independent trials with the probability of success as p, are [Set-02]

A. $\sqrt{np}$, $np(1-2p)$ **B.** np, np

C. np, $np(1-p)$ **D.** $\sqrt{np}$, $\sqrt{np(1-p)}$

Q3.20.1 NAT 2020 1M

A fair coin is tossed 20 times. The probability that 'head' will appear exactly 4 times in the first ten tosses, and 'tail' will appear exactly 4 times in the next ten tosses is _____ (round off to 3 decimal places)

Q3.20.2 NAT 2020 1M

A company is hiring to fill four managerial vacancies. The candidates are five men and three women. If every candidate is equally likely to be chosen then the probability that at least one women will be selected is _____ (round off to 2 decimal places).

Q3.20.3 NAT 2020 2M

Consider two exponentially distributed random variables X and Y, both having a mean of 0.50. Let $Z = X + Y$ and r be the correlation coefficient between X and Y. If the variance of Z equals 0, then the value of r is _____ (round off to 2 decimal places).

Q3.19.1 MCQ 2019 2M

The variable X takes a value between 0 and 10 with uniform probability distribution. The variable Y takes a value between 0 and 20 with uniform probability distribution. The probability of the sum of variable $(X + Y)$ being greater than 20 is

A. 0.25 **B.** 0.5 **C.** 0 **D.** 0.33

Q3.19.2 NAT 2019 1M

If x is the mean of data 3, x, 2, 4 then the mode is _____.

Q3.19.3 NAT 2019 2M

The probability that a part manufactured by a company will be defective is 0.05. If 15 such parts are selected randomly and inspected, then the probability that atleast two parts will be defective is _____.

Q3.18.1 MCQ 2018 1M

Four red balls, four green balls and four blue balls are put in a box. Three balls are pulled out of the box at random one after another without replacement. The probability that all the three balls are red is

A. $\frac{1}{72}$ **B.** $\frac{1}{55}$ **C.** $\frac{1}{36}$ **D.** $\frac{1}{27}$

Q3.18.2 MCQ 2018 1M

A six-faced fair dice is rolled five times. Then probability (in %) of obtaining 'ONE' at least four times is

A. 33.3 **B.** 3.33 **C.** 0.33 **D.** 0.0033

Q3.18.3 MCQ 2018 2M

Let X_1, X_2 be two normal random variable with means μ_1, μ_2 and standard deviation σ_1, σ_2 respectively. Consider $Y = X_1 - X_2$; $\mu_1 = \mu_2 = 1$, $\sigma_1 = 1$, $\sigma_2 = 2$ then,

A. Y is normal distributed random with mean 0 and variance 1.

B. Y is normal distributed with mean 0 and variance 5.

C. Y has mean 0 and variance 5, but is NOT normally distributed.

D. Y has mean 0 and variance 1, but is NOT normally distributed.

Q3.18.4 MCQ 2018 2M

Let X_1 and X_2 be two independent exponentially distributed random variables with means 0.5 and 0.25 respectively. Then $Y = \min(X_1, X_2)$ is

A. Exponentially distributed with mean 1/6.

B. Exponentially distributed with mean 2.

C. Normally distributed with mean 3/4.

D. Normally distributed with mean 1/6.

Q3.17.1 NAT 2017 1M

Two coins are tossed simultaneously. The probability (upto two decimal points accuracy) of getting at least one head is _____.

Q3.17.2 MCQ 2017 1M

A sample of 15 data is as follows: 17, 18, 17, 17, 13, 18, 5, 5, 6, 7, 8, 9, 20, 17, 3. The mode of the data is

A. 4 **B.** 13 **C.** 17 **D.** 20

Q3.17.3 NAT 2017 1M

A six-face fair dice is rolled a large number of times. The mean value of the outcomes is _____.

Q3.16.1 MCQ 2016 1M

Consider a Poisson distribution for the tossing of a biased coin. The mean for this distribution is μ. The standard deviation for this distribution is given by

A. $\sqrt{\mu}$ **B.** μ^2 **C.** μ **D.** $\frac{1}{\mu}$

Q3.16.2 NAT 2016 2M

The probability that a screw manufactured by a company is defective is 0.1. The company sells screws in packets containing 5 screws and gives a guarantee of replacement if one or more screws in the packet are found to be defective. The probability that a packet would have to be replaced is _____.

Q3.16.3 NAT 2016 1M

The area (in percentage) under standard normal distribution curve of random variable Z within limits from -3 to $+3$ is _____.

Q3.16.4 MCQ 2016 1M

Three cards were drawn from a pack of 52 cards. The probability that they are a king, a queen, and a jack is

A. $\frac{16}{5525}$ **B.** $\frac{64}{2197}$ **C.** $\frac{3}{13}$ **D.** $\frac{8}{16575}$

Q3.15.1 MCQ 2015 2M

The probability of obtaining at least two "SIX" in throwing a fair dice 4 times is

A. $\frac{425}{432}$ **B.** $\frac{19}{144}$ **C.** $\frac{13}{144}$ **D.** $\frac{125}{432}$

Q3.15.2 MCQ 2015 1M

If $P(X) = \frac{1}{4}$, $P(Y) = \frac{1}{3}$, and $P(X \cap Y) = \frac{1}{12}$, the value of $P(Y|X)$ is

A. $\frac{1}{4}$ **B.** $\frac{4}{25}$ **C.** $\frac{1}{3}$ **D.** $\frac{29}{50}$

Q3.15.3 MCQ 2015 1M

Among the four normal distribution with probability density functions as shown below, which one has the lowest variance?

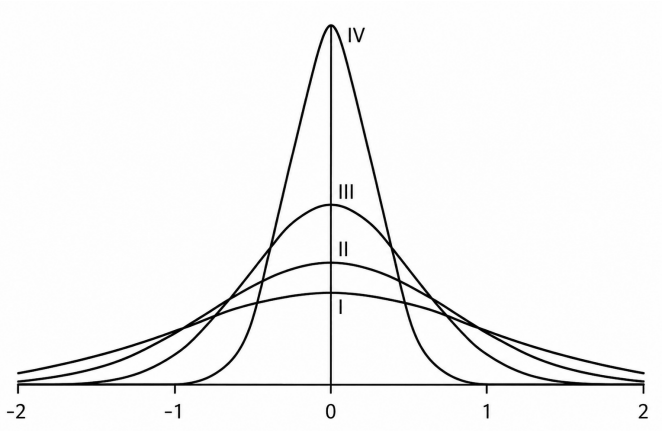


Fig. Q3.15.3

A. I **B.** II **C.** III **D.** IV

Q3.15.4 NAT 2015 1M

Three vendors were asked to supply a very high precision component. The respective probabilities of their meeting the strict design specifications are 0.8, 0.7 and 0.5. Each vendor supplies one component. The probability that out of total three components supplied by the vendors, at least one will meet the design specification is _____.

Q3.15.5 MCQ 2015 2M

The chance of a student passing an exam is 20%. The chance of a student passing the exam and getting above 90% marks in it is 5%. Given that a student passes the examination, the probability that the student gets above 90% marks is

A. $\frac{1}{18}$ **B.** $\frac{1}{4}$ **C.** $\frac{2}{9}$ **D.** $\frac{5}{18}$

Q3.14.1 MCQ 2014 2M

In the following table, x is a discrete random variable and $p(x)$ is the probability density. The standard deviation of x is

x	1	2	3
$p(x)$	0.3	0.6	0.1

A. 0.18 **B.** 0.36 **C.** 0.54 **D.** 0.6

Q3.14.2 MCQ 2014 2M

A machine produces 0, 1 or 2 defective pieces in a day with associated probability of $\frac{1}{6}$, $\frac{2}{3}$ and $\frac{1}{6}$, respectively. The mean value and the variance of the number of defective pieces produced by the machine in a day, respectively, are

A. 1 and $\frac{1}{3}$ **B.** $\frac{1}{3}$ and 1

C. 1 and $\frac{4}{3}$ **D.** $\frac{1}{3}$ and $\frac{4}{3}$

Q3.14.3 NAT 2014 2M

Consider an unbiased cubic dice with opposite faces coloured identically and each face coloured red, blue or green such that each colour appears only two times on the dice. If the dice is thrown thrice, the probability of obtaining red colour on top face of the dice at least two times is _____.

Q3.14.4 MCQ 2014 2M

The number of accidents occurring in a plant in a month follows Poisson distribution with mean as 5.2. The probability of occurrence of less than 2 accidents in the plant during a randomly selected month is

A. 0.029 **B.** 0.034 **C.** 0.039 **D.** 0.044

Q3.14.5 MCQ 2014 1M

A box contains 25 parts of which 10 are defective. Two parts are being drawn simultaneously in a random manner from the box. The probability of both the parts being good is

A. $\frac{7}{20}$ **B.** $\frac{42}{125}$ **C.** $\frac{25}{29}$ **D.** $\frac{5}{9}$

Q3.14.6 NAT 2014 2M

A group consists of equal number of men and women. Of this group 20% of the men and 50% of the women are unemployed. If a person is selected at random from this group, the probability of the selected person being employed is _____.

Q3.14.7 NAT 2014 1M

Demand during lead time with associated probabilities is shown below

Demand	50	70	75	80	85
Probability	0.15	0.14	0.21	0.20	0.30

Expected demand during lead time is _____

Q3.14.8 NAT 2014 2M

A nationalized bank has found that the daily balance available in its savings accounts follows a normal distribution with a mean of Rs. 500 and a standard deviation of Rs. 50. The percentage of savings account holders, who maintain an average daily balance more than Rs. 500 is _____

Q3.14.9 NAT 2014 1M

A batch of one hundred bulbs is inspected by testing four randomly chosen bulbs. The batch is rejected if even one of the bulbs is defective. A batch typically has five defective bulbs. The probability that the current batch accepted is _____

Q3.13.1 MCQ 2013 1M

Let X be a normal random variable with mean 1 and variance 4. The probability $P\{X < 0\}$ is

A. 0.5

B. greater than zero and less than 0.5

C. greater than 0.5 and less than 1.0

D. 1.0

Q3.13.2 MCQ 2013 2M

Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7?

A. $\frac{13}{90}$ **B.** $\frac{12}{90}$ **C.** $\frac{78}{90}$ **D.** $\frac{77}{90}$

Q3.13.3 MCQ 2013 2M

The probability that a student knows the correct answer to a multiple choice question is $\frac{2}{3}$. If the student does not know the answer, then the student guesses the answer. The probability of the guessed answer being correct is $\frac{1}{4}$. Given that the student has answered the question correctly, the conditional probability that the student knows the correct answer is

A. $\frac{2}{3}$ **B.** $\frac{3}{4}$ **C.** $\frac{5}{6}$ **D.** $\frac{8}{9}$

Q3.12.1 MCQ 2012 2M

A box contains 4 red balls and 6 black balls. Three balls are selected randomly from the box one after another, without replacement. The probability that the selected set contains one red ball and two black balls is

A. $\frac{1}{20}$ **B.** $\frac{1}{12}$ **C.** $\frac{3}{10}$ **D.** $\frac{1}{2}$

Q3.12.2 MCQ 2012 2M

An automobile plant contracted to buy shock absorbers from two suppliers X and Y. X supplies 60% and Y supplies 40% of the shock absorbers. All shock absorbers are subjected to a quality test. The ones that pass the quality test are considered reliable. Of X's shock absorbers, 96% are reliable. Of Y's shock absorbers, 72% are reliable.The probability that a randomly chosen shock absorber, which is found to be reliable, is made by Y is

A. 0.288 **B.** 0.334 **C.** 0.667 **D.** 0.720

Q3.11.1 MCQ 2011 2M

An unbiased coin is tossed five times. The outcome of each toss is either a head or a tail. The probability of getting at least one head is

A. $\frac{1}{32}$ **B.** $\frac{13}{32}$ **C.** $\frac{16}{32}$ **D.** $\frac{31}{32}$

Q3.10.1 MCQ 2010 2M

A box contains 2 washers, 3 nuts and 4 bolts. Items are drawn from the box at random one at a time without replacement. The probability of drawing 2 washers first followed by 3 nuts and subsequently the 4 bolts is

A. $\frac{2}{315}$ **B.** $\frac{1}{630}$ **C.** $\frac{1}{1260}$ **D.** $\frac{1}{2520}$

Q3.09.1 MCQ 2009 1M

If three coins are tossed simultaneously, the probability of getting at least one head is

A. $\frac{1}{8}$ **B.** $\frac{3}{8}$ **C.** $\frac{1}{2}$ **D.** $\frac{7}{8}$

Q3.09.2 MCQ 2009 2M

The standard deviation of a uniformly distributed random variable between 0 and 1 is

A. $\frac{1}{\sqrt{12}}$ **B.** $\frac{1}{\sqrt{3}}$ **C.** $\frac{5}{\sqrt{12}}$ **D.** $\frac{7}{\sqrt{12}}$

Q3.08.1 MCQ 2008 1M

A coin is tossed 4 times. What is the probability of getting heads exactly 3 times?

A. $\frac{1}{4}$ **B.** $\frac{3}{8}$ **C.** $\frac{1}{2}$ **D.** $\frac{3}{4}$

Q3.07.1 MCQ 2007 2M

Let X and Y be two independent random variables. Which one of the relations between expectation (E), variance (Var) and covariance (Cov) given below is False?

A. $E(XY) = E(X)\,E(Y)$

B. $\text{Cov}(X, Y) = 0$

C. $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

D. $E(X^2Y^2) = (E(X))^2\,(E(Y))^2$

Q3.06.1 MCQ 2006 1M

A box contains 20 defective items and 80 non-defective items. If two items are selected at random without replacement, what will be the probability that both the items are defective?

A. $\frac{1}{5}$ **B.** $\frac{1}{25}$ **C.** $\frac{20}{99}$ **D.** $\frac{19}{495}$

Q3.06.2 MCQ 2006 2M

Consider the continuous random variable with probability density function

$$f(t) = 1 + t \quad \text{for } -1 \le t \le 0$$

$$= 1 - t \quad \text{for } 0 \le t \le 1$$

The standard deviation of the random variable is

A. $\frac{1}{\sqrt{3}}$ **B.** $\frac{1}{\sqrt{6}}$ **C.** $\frac{1}{3}$ **D.** $\frac{1}{6}$

Q3.05.1 MCQ 2005 1M

A lot has 10% defective items. Ten items are chosen randomly from this lot. The probability that exactly 2 of the chosen items are defective is

A. 0.0036 **B.** 0.1937 **C.** 0.2234 **D.** 0.3874

Q3.05.2 MCQ 2005 2M

A single die is thrown twice. What is the probability that the sum is neither 8 nor 9?

A. $\frac{1}{9}$ **B.** $\frac{5}{36}$ **C.** $\frac{1}{4}$ **D.** $\frac{3}{4}$

Q3.04.1 MCQ 2004 2M

The following data about the flow of liquid was observed in a continuous chemical process plant

Flow rate (litres/sec)	7.5 to 7.7	7.7 to 7.9	7.9 to 8.1	8.1 to 8.3	8.3 to 8.5	8.5 to 8.7
Frequency	1	5	35	17	12	10

Mean flow rate of the liquid is

A. 8.00 litres/sec **B.** 8.06 litres/sec

C. 8.16 litres/sec **D.** 8.26 litres/sec

Q3.04.2 MCQ 2004 2M

From a pack of regular playing cards, two cards are drawn at random. What is the probability that both cards will be Kings, if first card is NOT replaced?

A. $\frac{1}{26}$ **B.** $\frac{1}{52}$ **C.** $\frac{1}{169}$ **D.** $\frac{1}{221}$

Q3.03.1 MCQ 2003 2M

A box contains 5 black and 5 red balls. Two balls are randomly picked one after another from the box, without replacement. The probability for both balls being red is

A. $\frac{1}{90}$ **B.** $\frac{1}{2}$ **C.** $\frac{19}{90}$ **D.** $\frac{2}{9}$

Q3.03.2 MCQ 2003 2M

Let $P(E)$ denote the probability of an event E. Given $P(A) = 1$, $P(B) = \frac{1}{2}$, the values of $P\left(\frac{A}{B}\right)$ and $P\left(\frac{B}{A}\right)$ respectively are

A. $\frac{1}{4}, \frac{1}{2}$ **B.** $\frac{1}{2}, \frac{1}{4}$ **C.** $\frac{1}{2}, 1$ **D.** $1, \frac{1}{2}$

Q3.02.1 MCQ 2002 1M

Two dice are thrown. What is the probability that the sum of the numbers on the two dice is eight?

A. $\frac{5}{36}$ **B.** $\frac{5}{18}$ **C.** $\frac{1}{4}$ **D.** $\frac{1}{3}$

Q3.02.2 MCQ 2002 2M

Manish has to travel from A to D changing buses at stops B and C enroute. The maximum waiting time at either stop can be 8 minutes each, but any time of waiting up to 8 minutes is equally likely at both places. He can afford up to 13 minutes of total waiting time if he is to arrive at D on time. What is the probability that Manish will arrive late at D?

A. $\frac{8}{13}$ **B.** $\frac{13}{64}$ **C.** $\frac{119}{128}$ **D.** $\frac{9}{128}$

Q3.02.3 MCQ 2002 2M

Arrivals at a telephone booth are considered to be Poisson, with an average time of 10 minutes between successive arrivals. The length of a phone call is distributed exponentially with mean 3 minutes. The probability that an arrival does not have to wait before service is

A. 0.3 **B.** 0.5 **C.** 0.7 **D.** 0.9

Q3.01.1 MCQ 2001 2M

An unbiased coin is tossed three times. The probability that the head turns up in exactly two cases is

A. $\frac{1}{9}$ **B.** $\frac{1}{8}$ **C.** $\frac{2}{3}$ **D.** $\frac{3}{8}$

Q3.00.1 MCQ 2000 1M

E_1 and E_2 are events in a probability space satisfying the following constants $P(E_1) = P(E_2)$, $P(E_1 \cup E_2) = 1$, E_1 and E_2 are independent then $P(E_1) =$ _____.

A. 0 **B.** $\frac{1}{4}$ **C.** $\frac{1}{2}$ **D.** 1

Q3.00.2 MCQ 2000 2M

In a manufacturing plant, the probability of making a defective bolt is 0.1. The mean and standard deviation of defective bolts in a total of 900 bolts are respectively

A. 90 and 9 **B.** 9 and 90

C. 18 and 9 **D.** 9 and 81

Q3.99.1 MCQ 1999 2M

At a production machine, parts arrive according to a Poisson process at the rate of 0.35 parts per minute. Processing time for parts have exponential distribution with mean of 2 minutes. What is the probability that a random part arrival finds that there are already 8 parts in the system (in machine + in queue)?

A. 0.0247 **B.** 0.0576 **C.** 0.0173 **D.** 0.082

Q3.98.1 MCQ 1998 1M

The probability that two friends share the same birth-month is:

A. $\frac{1}{6}$ **B.** $\frac{1}{12}$ **C.** $\frac{1}{144}$ **D.** $\frac{1}{24}$

Q3.97.1 MCQ 1997 2M

A box contains 5 black balls and 3 red balls. A total of three balls are picked from the box one after another, without replacing them back. The probability of getting two black balls and one red ball is

A. $\frac{3}{8}$ **B.** $\frac{2}{15}$ **C.** $\frac{15}{28}$ **D.** $\frac{1}{2}$

Q3.96.1 MCQ 1996 2M

The probability of a defective piece being produced in a manufacturing process is 0.01. The probability that out of 5 successive pieces, only one is defective, is

A. $(0.99)^4(0.01)$ **B.** $(0.99)(0.01)^4$

C. $5 \times (0.99)(0.01)^4$ **D.** $5 \times (0.99)^4(0.01)$

Q3.93.1 MCQ 1993 1M

If 20 percent managers are technocrats, the probability that a random committee of 5 managers consists of exactly 2 technocrats is:

A. 0.2048 **B.** 0.4000

C. 0.4096 **D.** 0.9421

Civil Engineering

Q3.26.1 NAT 2026 1M

Bag I contains 4 white and 6 black balls.
Bag II contains 4 white and 3 black balls.
One ball is drawn at random from any one of the two bags and it is found to be a black ball. The probability that the black ball was drawn from Bag I is _____ (rounded off to two decimal places).

Q3.26.2 NAT 2026 2M

The age (in years) of a population is normally distributed with a mean of 36 and standard deviation of 12. The height (in cm) of the same population is also normally distributed with a mean of 160 and standard deviation of 10.
If the probability of age greater than 50 years is equal to the probability of height greater than h, the value of h (in cm) is _____ (rounded off to two decimal places).

Q3.26.3 NAT 2026 1M

The probability (in %) that a storm having return period of 15 years may occur in the next 10 years is _____ (rounded off to two decimal places).

Q3.25.1 NAT 2025 2M

A one-way, single lane road has traffic that consists of 30% trucks and 70% cars. The speed of trucks (in km/h) is a uniform random variable on the interval $(30, 60)$, and the speed of cars (in km/h) is a uniform random variable on the interval $(40, 80)$. The speed limit on the road is 50 km/h. The percentage of vehicles that exceed the speed limit is _____ (rounded off to 1 decimal place).

Note: X is a uniform random variable on the interval (α, β), if its probability density function is given by

$$f(x) = \begin{cases} \frac{1}{\beta - \alpha} & \alpha < x < \beta \\ 0 & \text{otherwise} \end{cases}$$

Q3.25.2 MCQ 2025 1M

X is a random variable that can take any one of the values 0, 1, 7, 11, and 12. The probability mass function for X is $P(X = 0) = 0.4$; $P(X = 1) = 0.3$; $P(X = 7) = 0.1$; $P(X = 11) = 0.1$; $P(X = 12) = 0.1$
Then, the variance of X is

A. 20.81 **B.** 28.40 **C.** 31.70 **D.** 10.89

Q3.25.3 MSQ 2025 1M

Given that A and B are not null sets, which of the following statements regarding probability is/are CORRECT?

☐ **A.** $P(A \cap B) = P(A)\,P(B)$, if A and B are mutually exclusive.

☐ **B.** Conditional probability, $P(A|B) = 1$ if $B \subset A$.

☐ **C.** $P(A \cup B) = P(A) + P(B)$, if A and B are mutually exclusive.

☐ **D.** $P(A \cap B) = 0$, if A and B are independent.

Q3.25.4 NAT 2025 2M

Consider a discrete random variable X whose probabilities are given below. The standard deviation of the random variable is _____ (round off to one decimal place).

x_i	1	2	4	8
$P(X = x_i)$	0.3	0.1	0.3	0.3

Q3.24.1 MCQ 2024 2M

In a sample of 100 heart patients, each patient has 80% chance of having a heart attack without medicine X. It is clinically known that medicine X reduces the probability of having a heart attack by 50%. Medicine X is taken by 50 of these 100 patients. The probability that a randomly selected patient, out of the 100 patients, takes medicine X and has a heart attack is

A. 40% **B.** 60% **C.** 20% **D.** 30%

Q3.24.2 MCQ 2024 1M

The probability that a student passes only in Mathematics is $\frac{1}{3}$. The probability that the student passes only in English is $\frac{4}{9}$. The probability that the student passes in both of these subjects is $\frac{1}{6}$. The probability that the student will pass in at least one of these two subjects is

A. $\frac{17}{18}$ **B.** $\frac{11}{18}$ **C.** $\frac{14}{18}$ **D.** $\frac{1}{18}$

Q3.23.1 MCQ 2023 1M

A remote village has exactly 1000 vehicles with sequential registration numbers starting from 1000. Out of the total vehicles, 30% are without pollution clearance certificate. Further, even- and odd-numbered vehicles are operated on even- and odd-numbered dates, respectively.

If 100 vehicles are chosen at random on an even-numbered date, the number of vehicles expected without pollution clearance certificate is _____.

A. 15 **B.** 30 **C.** 50 **D.** 70

Q3.23.2 NAT 2023 1M

The probabilities of occurrences of two independent events A and B are 0.5 and 0.8, respectively. What is the probability of occurrence of at least A or B (rounded off to one decimal place)? _____

Q3.23.3 MCQ 2023 1M

Which of the following probability distribution functions (PDFs) has the mean greater than the median?

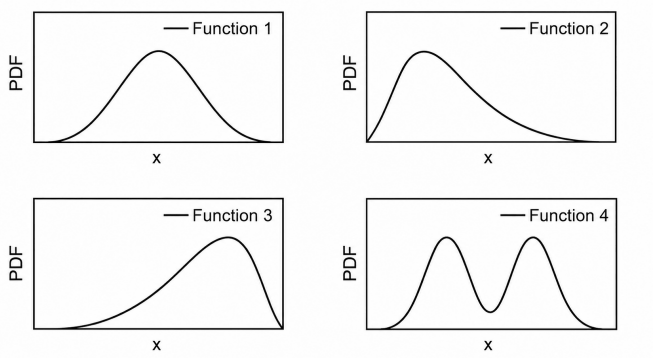


Fig. Q3.23.3

A. Function 1 **B.** Function 2
C. Function 3 **D.** Function 4

Q3.22.1 NAT 2022 2M

At a traffic intersection, cars and buses arrive randomly according to independent Poisson processes at an average rate of 4 vehicles per hour and 2 vehicles per hour, respectively. The probability of observing at least 2 vehicles in 30 minutes is _____. (*round off to two decimal places*)

Q3.22.2 NAT 2022 2M

A pair of six-faced dice is rolled thrice. The probability that the sum of the outcomes in each roll equals 4 in exactly two of the three attempts is _____. (*round off to three decimal places*)

Q3.22.3 NAT 2022 2M

A set of observations of independent variable (x) and the corresponding dependent variable (y) is given below.

x	5	2	4	3
y	16	10	13	12

Based on the data, the coefficient a of the linear regression model

$$y = a + bx$$

is estimated as 6.1.

The coefficient b is _____. (*round off to one decimal place*)

Q3.21.1 NAT 2021 2M

Vehicular arrival at an isolated intersection follows the Poisson distribution. The mean of vehicular arrival rate is 2 vehicle per minute. The probability (*round off to two decimal places*) that at least 2 vehicles will arrive in any given 1-minute interval is _____

Q3.21.2 MCQ 2021 1M

The shape of the cumulative distribution function of Gaussian distribution is

A. Bell-shaped

B. S-shaped

C. Horizontal line

D. Straight line at 45 degree angle

Q3.20.1 NAT 2020 1M

The probability that a 50 year flood may NOT occur at all during 25 years life of a project (*round off to two decimal places*), is _____.

Q3.20.2 NAT 2020 1M

A fair (unbiased) coin is tossed 15 times. The probability of getting exactly 8 Heads (*round off to three decimal places*), is _____.

Q3.19.1 MCQ 2019 2M

The probability density function of a continuous random variable distributed uniformly between x and y (for $y > x$) is

A. $\frac{1}{x-y}$ **B.** $\frac{1}{y-x}$ **C.** $x-y$ **D.** $y-x$

Q3.19.2 NAT 2019 2M

Traffic on a highway is moving at a rate of 360 vehicles per hour at a location. If the number of vehicles arriving on this highway follows Poisson distribution, the probability (round off to 2 decimal places) that the headway between successive vehicles lies between 6 and 10 seconds is _____.

Q3.18.1 NAT 2018 1M

Probability (up to one decimal place) of consecutively picking 3 red balls without replacement from a box containing 5 red balls and 1 white ball is _____.

Q3.18.2 MCQ 2018 1M

A probability distribution with right skew is shown in the figure.

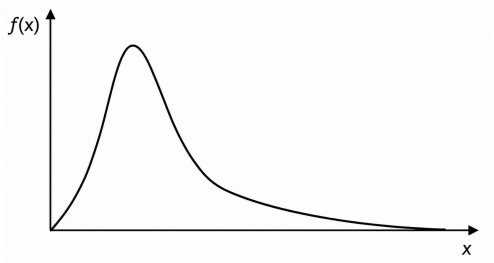


Fig. Q3.18.2

The correct statement for the probability distribution is

A. Mean is equal to mode

B. Mean is greater than median but less than mode

C. Mean is greater than median and mode

D. Mode is greater than median

Q3.18.3 MCQ 2018 1M

The graph of a function $f(x)$ is shown in the figure.

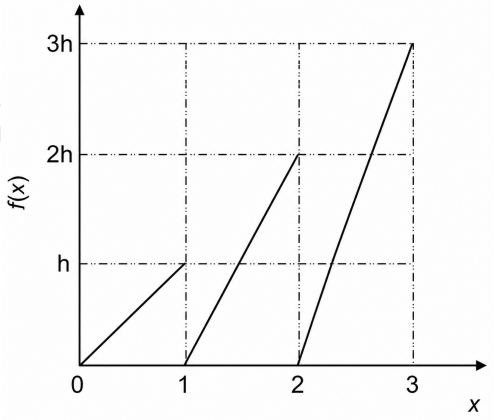


Fig. Q3.18.3

For $f(x)$ to be a valid probability density function, the value of h is

A. $1/3$ **B.** $2/3$ **C.** 1 **D.** 3

Q3.18.4 NAT 2018 1M

The frequency distribution of the compressive strength of 20 concrete cube specimens is given in the table.

f (MPa)	Number of specimens with compressive strength equal to f
23	4
28	2
22.5	5
31	5
29	4

If μ is the mean strength of the specimens and σ is the standard deviation, the number of specimens (out of 20) with compressive strength less than $\mu - 3\sigma$ is _____.

Q3.17.1 NAT 2017 1M

For a construction project, the mean and standard deviation of the completion time are 200 days and 6.1 days, respectively. Assume normal distribution and use the value of standard normal deviate $z = 1.64$ for the 95% confidence level. The maximum time required (in days) for the completion of the project would be _____.

Q3.17.2 NAT 2017 1M

A two-faced fair coin has its faces designated as head (H) and tail (T). This coin is tossed three times in succession to record the following outcomes: H, H, H. If the coin is tossed one more time, the probability (up to one decimal place) of obtaining H again, given the previous realizations of H, H and H, would be _____.

Q3.17.3 MCQ 2017 2M

For the function $f(x) = a + bx$, $0 \le x \le 1$, to be a valid probability density function, which one of the following statements is correct?

A. $a = 1,\ b = 4$ **B.** $a = 0.5,\ b = 1$

C. $a = 0,\ b = 1$ **D.** $a = 1,\ b = -1$

Q3.17.4 NAT 2017 1M

Vehicles arriving at an intersection from one of the approach road follow the Poisson distribution. The mean rate of arrival is 900 vehicles per hour. If a gap is defined as the time difference between two successive vehicle arrivals (with vehicles assumed to be points), the probability (up to four decimal places) that the gap is greater than 8 seconds is _____.

Q3.16.1 MCQ 2016 2M

If $f(x)$ and $g(x)$ are two probability density functions,

$$f(x) = \begin{cases} \frac{x}{a} + 1, & -a \le x < 0 \\ -\frac{x}{a} + 1, & 0 \le x \le a \\ 0, & \text{otherwise} \end{cases}$$

$$g(x) = \begin{cases} -\frac{x}{a}, & -a \le x < 0 \\ \frac{x}{a}, & 0 \le x \le a \\ 0, & \text{otherwise} \end{cases}$$

Which one of the following statements is true?

A. Mean of $f(x)$ and $g(x)$ are same; Variance of $f(x)$ and $g(x)$ are same

B. Mean of $f(x)$ and $g(x)$ are same; Variance of $f(x)$ and $g(x)$ are different

C. Mean of $f(x)$ and $g(x)$ are different; Variance of $f(x)$ and $g(x)$ are same

D. Mean of $f(x)$ and $g(x)$ are different; Variance of $f(x)$ and $g(x)$ are different

Q3.16.2 MCQ 2016 2M

Probability density function of a random variable X is given below

$$f(x) = \begin{cases} 0.25, & \text{if } 1 \le x \le 5 \\ 0, & \text{otherwise} \end{cases}$$

$P(X \le 4)$ is

A. $\frac{3}{4}$ **B.** $\frac{1}{2}$ **C.** $\frac{1}{4}$ **D.** $\frac{1}{8}$

Q3.16.3 MCQ 2016 1M

X and Y are two random independent events. It is known that $P(X) = 0.40$ and $P(X \cup Y^c) = 0.7$. Which one of the following is the value of $P(X \cup Y)$?

A. 0.7 **B.** 0.5 **C.** 0.4 **D.** 0.3

Q3.16.4 NAT 2016 1M

The spot speeds (expressed in km/hr) observed at a road section are 66, 62, 45, 79, 32, 51, 56, 60, 53, and 49. The median speed (expressed in km/hr) is _____.

(Note: answer with one decimal accuracy)

Q3.15.1 NAT 2015 1M

Consider the following probability mass function (p.m.f.) of a random variable X:

$$p(x,q) = \begin{cases} q, & \text{if } X = 0, \\ 1-q, & \text{if } X = 1, \\ 0, & \text{otherwise.} \end{cases}$$

If $q = 0.4$, the variance of X is _____.

Q3.15.2 NAT 2015 2M

The probability density function of a random variable, x is

$$f(x) = \begin{cases} \frac{x}{4}(4-x^2), & \text{for } 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

The mean, μ_x of the random variable is _____.

Q3.14.1 NAT 2014 1M

The probability density function of evaporation E on any day during a year in a watershed is given by

$$f(E) = \begin{cases} \frac{1}{5} & 0 \le E \le 5 \text{ mm/day} \\ 0 & \text{otherwise} \end{cases}$$

The probability that E lies in between 2 and 4 mm/day in a day in the watershed is (in decimal) _____

Q3.14.2 NAT 2014 2M

An observer counts 240 veh/h at a specific highway location. Assume that the vehicle arrival at the location is Poisson distributed, the probability of having one vehicle arriving over a 30-second time interval is _____

Q3.14.3 MCQ 2014 1M

A fair (unbiased) coin was tossed four times in succession and resulted in the following outcomes: (i) Head, (ii) Head, (iii) Head, (iv) Head. The probability of obtaining a 'Tail' when the coin is tossed again is

A. 0 **B.** $\frac{1}{2}$ **C.** $\frac{4}{5}$ **D.** $\frac{1}{5}$

Q3.14.4 NAT 2014 2M

A traffic office imposes on an average 5 number of penalties daily on traffic violators. Assume that the number of penalties on different days is independent and follows a Poisson distribution. The probability that there will be less than 4 penalties in a day is _____

Q3.14.5 MCQ 2014 1M

If $\{x\}$ is a continuous, real valued random variable defined over the interval $(-\infty, +\infty)$ and its occurrence is defined by the density function given as:

$$f(x) = \frac{1}{\sqrt{2\pi}\, b} e^{-\frac{1}{2}\left(\frac{x-a}{b}\right)^2}$$

where 'a' and 'b' are the statistical attributes of the random variable $\{x\}$. The value of the integral

$$\int_{-\infty}^{a} \frac{1}{\sqrt{2\pi}\, b} e^{-\frac{1}{2}\left(\frac{x-a}{b}\right)^2} dx$$

is

A. 1 **B.** 0.5 **C.** π **D.** $\frac{\pi}{2}$

Q3.13.1 MCQ 2013 1M

A 1 hour rainfall of 10 cm has return period of 50 year. The probability that 1 hour of rainfall 10 cm or more will occur in each of two successive years is

A. 0.04 **B.** 0.2 **C.** 0.2 **D.** 0.0004

Q3.13.2 NAT 2013 2M

Find the value of λ such that function $f(x)$ is valid probability density function?

$$f(x) = \begin{cases} \lambda(x-1)(2-x) & \text{for } 1 \le x \le 2 \\ 0 & \text{otherwise} \end{cases}$$

Q3.12.1 MCQ 2012 2M

An automobile plant contracted to buy shock absorbers from two suppliers X and Y. X supplies 60% and Y supplies 40% of the shock absorbers. All shock absorbers are subjected to a quality test. The ones that pass the quality test are considered reliable. Of X's shock absorbers, 96% are reliable. Of Y's shock absorbers, 72% are reliable.
The probability that a randomly chosen shock absorber, which is found to be reliable, is made by Y is

A. 0.288 **B.** 0.334 **C.** 0.667 **D.** 0.720

Q3.12.2 MCQ 2012 2M

In an experiment, positive and negative values are equally likely to occur. The probability of obtaining atmost one negative value in five trials is

A. $\frac{1}{32}$ **B.** $\frac{2}{32}$ **C.** $\frac{3}{32}$ **D.** $\frac{6}{32}$

Q3.12.3 MCQ 2012 1M

The annual precipitation data of a city is normally distributed with mean and standard deviation as 1000mm and 200mm, respectively. The probability that the annual precipitation will be more than that of 1200mm is

A. < 50% **B.** 50% **C.** 75% **D.** 100%

Q3.11.1 MCQ 2011 1M

There are two containers, with one containing 4 Red and 3 Green balls and the other containing 3 blue and 4 Green balls. One ball is drawn at random from each container. The probability that one of the balls is Red and the other is Blue will be

A. $\frac{1}{7}$ **B.** $\frac{9}{49}$ **C.** $\frac{12}{49}$ **D.** $\frac{3}{7}$

Q3.11.2 MCQ 2011 1M

The probability that k number of vehicles arrive (i.e., cross a predefined line) in time t is given as $(\lambda t)^k e^{-\lambda t}/k!$. where λ is the average vehicle arrival rate. What is the probability that the time headway is greater than or equal to time t_1?

A. $\lambda e^{\lambda t_1}$ **B.** λe^{-t_1} **C.** $e^{\lambda t_1}$ **D.** $e^{-\lambda t_1}$

Q3.10.1 MCQ 2010 1M

Two coins are simultaneously tossed. The probability of two heads simultaneously appearing is

A. $\frac{1}{8}$ **B.** $\frac{1}{6}$ **C.** $\frac{1}{4}$ **D.** $\frac{1}{2}$

Q3.09.1 MCQ 2009 2M

The standard normal probability function can be approximated as

$$F(x_N) = \frac{1}{1 + \exp\left(-1.7255 x_N |x_N|^{0.12}\right)}$$

where x_N = standard normal deviate. If mean and standard deviation of annual precipitation are 102 cm and 27 cm respectively, the probability that the annual precipitation will be between 90 cm and 102 cm is

A. 66.7% **B.** 50.0% **C.** 33.3% **D.** 16.7%

Q3.08.1 MCQ 2008 2M

A person on a trip has a choice between private car and public transport. The probability of using a private car is 0.45. While using the public transport, further choices available are bus and metro, out of which the probability of commuting by a bus is 0.55. In such a situation, the probability (rounded up to two decimals) of using a car, bus and metro, respectively would be

A. 0.45, 0.30 and 0.25 **B.** 0.45, 0.25 and 0.30

C. 0.45, 0.55 and 0.00 **D.** 0.45, 0.35 and 0.20

Q3.08.2 MCQ 2008 2M

If probability density function of a random variable X is

$$f(x) = x^2 \text{ for } -1 \le x \le 1, \text{ and}$$

$$= 0 \text{ for any other value of } x$$

Then, the percentage probability

$$P\left(-\frac{1}{3} \le x \le \frac{1}{3}\right)$$

is

A. 0.247 **B.** 2.47 **C.** 24.7 **D.** 247

Q3.08.3 MCQ 2008 2M

Three values of x and y are to be fitted in a straight line in the form $y = a + bx$ by the method of least squares. Given: $\sum x = 6$, $\sum y = 21$, $\sum x^2 = 14$ and $\sum xy = 46$, the values of a and b are respectively

A. 2 and 3 **B.** 1 and 2

C. 2 and 1 **D.** 3 and 2

Q3.07.1 MCQ 2007 2M

If the standard deviation of the spot speed of vehicles in a highway is 8.8 kmph and the mean speed of the vehicles is 33 kmph, the coefficient of variation in speed is

A. 0.1517 **B.** 0.1867 **C.** 0.2666 **D.** 0.3646

Q3.06.1 MCQ 2006 2M

There are 25 calculators in a box. Two of them are defective. Suppose 5 calculators are randomly picked for inspection (i.e., each has the same chance of being selected), what is the probability that only one of the defective calculators will be included in the inspection?

A. $\frac{1}{2}$ **B.** $\frac{1}{3}$ **C.** $\frac{1}{4}$ **D.** $\frac{1}{5}$

Q3.06.2 MCQ 2006 2M

A class of first year B. Tech. students is composed of four batches A, B, C and D, each consisting of 30 students. It is found that the sessional marks of students in Engineering Drawing in batch C have a mean of 6.6 and standard deviation of 2.3. The mean and standard deviation for the entire class are 5.5 and 4.2, respectively. It is decided by the course instructor to normalize the marks of the students of all batches to have the same mean and standard deviation as that of the entire class. Due to this, the marks of a student in batch C are changed from 8.5 to

A. 6.0 **B.** 7.0 **C.** 8.0 **D.** 9.0

Q3.05.1 MCQ 2005 1M

Which one of the following statement is NOT true?

A. The measure of skewness is dependent upon the amount of dispersion.

B. In a symmetric distribution, the values of mean, mode and median are the same.

C. In a positively skewed distribution: mean > median > mode.

D. In a negatively skewed distribution: mode > mean > median.

Q3.04.1 MCQ 2004 2M

A hydraulic structure has four gates which operate independently. The probability of failure of each gate is 0.2. Given that gate 1 has failed, the probability that both gates 2 and 3 will fail is

A. 0.240 **B.** 0.200 **C.** 0.040 **D.** 0.008

Q3.03.1 MCQ 2003 1M

A box contains 10 screws, 3 of which are defective. Two screws are drawn at random with replacement. The probability that none of the two screws is defective will be

A. 100% **B.** 50%

C. 49% **D.** None of these

Chemical Engineering

Q3.26.1 NAT 2026 1M

An experiment is performed by rolling an unbiased die once. Events A and B are defined as follows:
Event A: The outcome of the rolled die is a prime number.
Event B: The outcome of the rolled die is an odd number less than 4.
The conditional probability of A given B, denoted by $P(A \mid B)$, is _____ (rounded off to one decimal place).

Q3.25.1 MCQ 2025 1M

A box contains 3 identical green balls and 7 identical blue balls. Two balls are randomly drawn without replacement from the box. The probability of drawing 1 green and 1 blue ball is

A. $\frac{{}^3P_1 \times {}^7P_1}{{}^{10}P_2}$ **B.** $\frac{{}^{10}P_3 \times {}^{10}P_7}{{}^{10}P_2}$

C. $\frac{{}^{10}C_3 \times {}^{10}C_7}{{}^{10}C_2}$ **D.** $\frac{{}^3C_1 \times {}^7C_1}{{}^{10}C_2}$

Q3.25.2 MCQ 2025 2M

A probability distribution function is given as

$$p(x) = \begin{cases} \frac{1}{a}, & x \in (0, a) \\ 0, & \text{otherwise} \end{cases}$$

where a is a positive constant. For a function $f(x) = x^2$, the expectation of $f(x)$ is

A. $\frac{a^2}{3}$ **B.** $\frac{a^3}{3}$ **C.** $\frac{2a^2}{3}$ **D.** $\frac{2a^3}{3}$

Q3.24.1 MCQ 2024 1M

Consider the normal probability distribution function

$$f(x) = \frac{4}{\sqrt{2\pi}} e^{-8(x+3)^2}$$

If μ and σ are the mean and standard deviation of $f(x)$ respectively, then the ordered pair (μ, σ) is

A. $\left(3, \frac{1}{4}\right)$ **B.** $\left(-3, \frac{1}{4}\right)$ **C.** $(3, 4)$ **D.** $(-3, 4)$

Q3.23.1 MCQ 2023 2M

An exhibition was held in a hall on 15 August 2022 between 3 PM and 4 PM during which any person was allowed to enter only once. Visitors who entered before 3:40 PM exited the hall exactly after 20 minutes from their time of entry. Visitors who entered at or after 3:40 PM, exited exactly at 4 PM. The probability distribution of the arrival time of any visitor is uniform between 3 PM and 4 PM. Two persons X and Y entered the exhibition hall independent of each other. Which one of the following values is the probability that their visits to the exhibition overlapped with each other?

A. $\frac{5}{9}$ **B.** $\frac{4}{9}$ **C.** $\frac{2}{9}$ **D.** $\frac{7}{9}$

Q3.21.1 NAT 2021 2M

The probability distribution function of a random variable X is shown in the following figure.

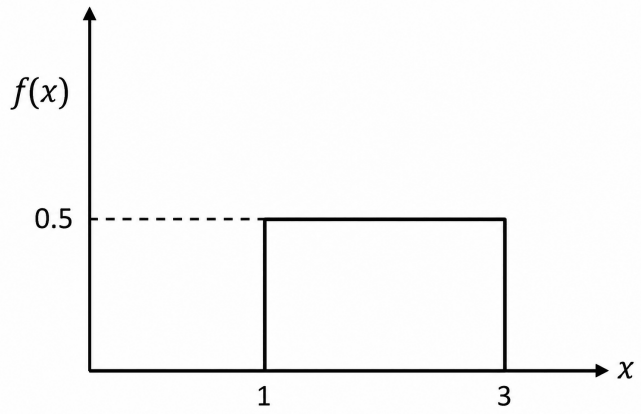


Fig. Q3.21.1

From this distribution, random samples with sample size $n = 68$ are taken. If $\overline{X}$ is the sample mean, the standard deviation of the probability distribution of $\overline{X}$, i.e. $\sigma_{\overline{X}}$ is _____ (round off to 3 decimal places).

Q3.20.1 MCQ 2020 2M

In a box, there are 5 green balls and 10 blue balls. A person picks 6 balls randomly. The probability that the person has picked 4 green balls and 2 blue balls is

A. $\frac{42}{1001}$ **B.** $\frac{45}{1001}$ **C.** $\frac{240}{1001}$ **D.** $\frac{420}{1001}$

Q3.19.1 NAT 2019 2M

Two unbiased dice are thrown. Each dice can show any number between 1 and 6. The probability that the sum of the outcomes of the two dice is divisible by 4 is _____ (rounded off to two decimal places).

Q3.18.1 MCQ 2018 1M

A watch uses two electronic circuits (ECs). Each EC has a failure probability of 0.1 in one year of operation. Both ECs are required for functioning of the watch. The probability of the watch functioning for one year without failure is

A. 0.99 **B.** 0.90 **C.** 0.81 **D.** 0.80

Q3.17.1 MCQ 2017 1M

The marks obtained by a set of students are: 38, 84, 45, 70, 75, 60, 48.

The mean and median marks, respectively, are

A. 45 and 75 **B.** 55 and 48

C. 60 and 60 **D.** 60 and 70

Q3.17.2 NAT 2017 2M

A box has 6 red balls and 4 white balls. A ball is picked at random and replaced in the box, after which a second ball is picked.

The probability of both the balls being red, rounded to 2 decimal places, is _____.

Q3.16.1 NAT 2016 2M

The model $y = mx^2$ is to be fit to the data given below.

x	1	$\sqrt{2}$	$\sqrt{3}$
y	2	5	8

Using linear regression, the value (rounded off to the second decimal place) of m is _____.

Q3.14.1 NAT 2014 2M

In rolling of two fair dice, the outcome of an experiment is considered to be the sum of the numbers appearing on the dice. The probability is highest for the outcome of _____.

Q3.14.2 MCQ 2014 2M

Consider the following two normal distributions:

$$f_1(x) = \exp(-\pi x^2)$$

$$f_2(x) = \frac{1}{2\pi} \exp\left\{-\frac{1}{4\pi}(x^2 + 2x + 1)\right\}$$

If μ and σ denote the mean and standard deviation, respectively, then

A. $\mu_1 < \mu_2$ and $\sigma_1^2 < \sigma_2^2$ **B.** $\mu_1 < \mu_2$ and $\sigma_1^2 > \sigma_2^2$
C. $\mu_1 > \mu_2$ and $\sigma_1^2 < \sigma_2^2$ **D.** $\mu_1 > \mu_2$ and $\sigma_1^2 > \sigma_2^2$

Q3.13.1 NAT 2013 1M

For two rolls of a fair die, the probability of getting a 4 in the first roll and a number less than 4 in the second roll, up to 3 digits after the decimal point, is _____.

Q3.13.2 MCQ 2013 1M

The number of emails received on six consecutive days is 11, 9, 18, 18, 4 and 15, respectively. What are the median and the mode for these data?

A. 18 and 11, respectively **B.** 13 and 18, respectively
C. 13 and 12.5, respectively **D.** 12.5 and 18, respectively

Q3.12.1 MCQ 2012 1M

A box containing 10 identical compartments has 6 red balls and 2 blue balls. If each compartment can hold only one ball, then the number of different possible arrangements are

A. 1026 **B.** 1062 **C.** 1260 **D.** 1620

Q3.10.1 MCQ 2010 2M

X and Y are independent random variables. X follows a binomial distribution, with $N = 5$ and $p = \frac{1}{2}$. Y takes integer values 1 and 2, with equal probability. Then the probability that $X = Y$ is

A. $\frac{15}{64}$ **B.** $\frac{15}{32}$ **C.** $\frac{1}{2}$ **D.** $\frac{15}{16}$

Q3.10.2 MCQ 2010 2M

A box contains three red and two black balls. Four balls are removed from the box one by one, without replacement. The probability of the ball remaining in the box being red, is

A. $\frac{609}{625}$ **B.** $\frac{3}{5}$ **C.** $\frac{2}{5}$ **D.** $\frac{81}{625}$

Q3.09.1 MCQ 2009 2M

A fair die is rolled. Let R denote the event of obtaining a number less than or equal to 5 and S denote the event of obtaining an odd number. Then which ONE of the following about the probability (P) is TRUE?

A. $P(R \mid S) = 1$ **B.** $P(R \mid S) = 0$
C. $P(S \mid R) = 1$ **D.** $P(S \mid R) = 0$

Q3.08.1 MCQ 2008 2M

The Poisson distribution is given by

$$P(r) = \frac{m^r}{r!} \exp(-m).$$

The first moment about the origin for this distribution is

A. 0 **B.** m **C.** $\frac{1}{m}$ **D.** m^2

Q3.06.1 MSQ 2006 2M

A pair of fair dice is rolled three times. What is the probability that 10 (sum of the number on the two faces) will show up exactly once?

A. $\frac{121}{1728}$ **B.** $\frac{363}{1728}$ **C.** $\frac{121}{576}$ **D.** $\frac{363}{576}$

Q3.06.2 MCQ 2006 2M

A company purchased components from the firms P, Q, and R as shown in the table below.

Firm	Total number of components purchased	Number of components likely to be defective
P	1000	5
Q	2500	5
R	500	2

The components are stored together. One of the component is selected at random and found to be defective. What is the probability that it was supplied by firm R?

A. $\frac{1}{250}$ **B.** $\frac{1}{12}$ **C.** $\frac{1}{8}$ **D.** $\frac{1}{6}$

Q3.05.1 MCQ 2005 1M

Two bags contain ten coins each and the coins in each bag are numbered from 1 to 10. One coin is drawn at random from each bag. The probability that one of the coins has value 1, 2, 3, or 4, while the other has value 7, 8, 9 or 10 is

A. $\frac{2}{5}$ **B.** $\frac{4}{25}$ **C.** $\frac{8}{25}$ **D.** $\frac{16}{25}$

Q3.04.1 MCQ 2004 1M

A box contains three blue balls and four red balls. Another identical box contains two blue balls and five red balls. One ball is picked at random from one of the two boxes and it is red. The probability that it came from the first box is

A. $\frac{2}{3}$ **B.** $\frac{4}{9}$ **C.** $\frac{4}{7}$ **D.** $\frac{2}{7}$

Q3.02.1 MCQ 2002 1M

A fair die is rolled four times. Find the probability that six shows up twice.

A. $\frac{1}{2}$ **B.** $\frac{16}{325}$ **C.** $\frac{1}{36}$ **D.** $\frac{25}{216}$

Production & Industrial Engineering

Q3.26.1 NAT 2026 2M

The number of soldering defects that occur in a semiconductor device follows a discrete Poisson distribution with a probability mass function

$$p(x) = \frac{e^{-\lambda}\lambda^{x}}{x!}.$$

The average number of soldering defects per semiconductor device is 3. The probability that a randomly selected semiconductor device will have at least two soldering defects is _____ (rounded off to two decimal places).

Q3.26.2 MCQ 2026 2M

The function, $r(t)$, signifies the likelihood that a component has survived up to time t and fails in the next instant of time.

$$r(t) = \frac{f(t)}{1 - F(t)}$$

where $f(t)$ denotes the probability density function for time to failure with corresponding cumulative distribution function given by,

$$F(t) = 1 - e^{-at^{b}}, \qquad a > 0,\ b > 0$$

Which ONE of the following represents $r(t)$?

A. $a \times b \times t^{b-1} \times e^{-at^{b}}$

B. $a \times b \times t^{b-1}$

C. $1 - \left(a \times b \times t^{b-1}\right)$

D. $1 - \left(a \times b \times t^{b-1} \times e^{-at^{b}}\right)$

Q3.26.3 NAT 2026 1M

A box in a machine shop consists of 5 coated and 10 uncoated cutting inserts which are of otherwise similar characteristics. The inserts got mixed up randomly in the box. An operator has taken 4 of them at once without noticing the differences to mount on a 4-tooth face milling cutter that uses inserts.

The probability of the milling cutter having all uncoated inserts is _____ (rounded off to two decimal places).

Q3.25.1 MCQ 2025 1M

A bag contains 5 red, 7 green and 3 blue balls. Two balls are drawn at random from the bag one-by-one. The probability of the second drawn ball being red is

A. $\frac{1}{5}$ **B.** $\frac{1}{3}$ **C.** $\frac{2}{5}$ **D.** $\frac{2}{3}$

Q3.24.1 NAT 2024 2M

Seven cards numbered 1 to 7 are placed in a box. After thoroughly mixing all the cards, one card is drawn at random.

If it is known that the number on the card drawn is odd, then the probability that the number on the card drawn is greater than 4 is _____ %. (Answer in integer)

Q3.24.2 NAT 2024 1M

If X is a continuous random variable with the probability density function

$$f(x) = \begin{cases} \frac{K}{4}, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

then the value of K is _____. (Answer in integer)

Q3.23.1 MCQ 2023 1M

Three unbiased coins are tossed. Provided that at least two outcomes are tails, the probability of having all three outcomes as tails is

A. $\frac{1}{8}$ **B.** $\frac{1}{4}$ **C.** $\frac{1}{3}$ **D.** $\frac{1}{2}$

Q3.23.2 MCQ 2023 2M

Two cards are drawn one after the other from a regular deck of 52 playing cards without replacement. The probability that the drawn cards are of different suits is

A. $\frac{39}{51}$ **B.** $\frac{13}{52}$ **C.** $\frac{2}{52}$ **D.** $\frac{2}{51}$

Q3.22.1 NAT 2022 1M

If the interarrival time is exponential and 8 customers per hour arrive in a bank, then the probability of no arrival of customer during a period of 15 minutes is _____. [round off to two decimal places]

Q3.22.2 NAT 2022 1M

An operator manufactures 10 identical spur gears in a lot. One spur gear is defective in the lot. Three spur gears are drawn at random without replacement. The probability of getting all three gears as non-defective is _____. [round off to two decimal places]

Q3.21.1 NAT 2021 2M

The probability mass function $P(x)$ of a discrete random variable X is given by $P(x) = \frac{1}{2^x}$, where $x = 1, 2, \cdots, \infty$. The expected value of X is _____. *[in integer]*

Q3.21.2 NAT 2021 2M

The time to pass through a security screening at an airport follows an exponential distribution. The mean time to pass through the security screening is 15 minutes. To catch the flight, a passenger must clear the security screening within 15 minutes. The probability that the passenger will miss the flight is _____. *[round off to 3 decimal places]*

Q3.20.1 MCQ 2020 1M

If x is a random variable with the expected value of 5 and the variance of 1, then the expected value of x^2 is

A. 24 **B.** 25 **C.** 26 **D.** 36

Q3.20.2 NAT 2020 2M

If the probability density function of a random variable x is given by

$$f(x) = \begin{cases} \frac{kx^2}{2}, & -1 \le x \le 1, \\ 0, & \text{elsewhere,} \end{cases}$$

the value of k is _____.

Q3.19.1 NAT 2019 2M

A box contains 2 red and 3 black balls. Three balls are randomly chosen from the box and are placed in a bag. Then the probability that there are 1 red and 2 black balls in the bag, is _____.

Q3.18.1 NAT 2018 1M

The probabilities of occurrence of events F and G are $P(F) = 0.3$ and $P(G) = 0.4$, respectively. The probability that both events occur simultaneously is $P(F \cap G) = 0.2$. The probability of occurrence of at least one event $P(F \cup G)$ is _____.

Q3.18.2 NAT 2018 1M

Weights (in kg) of six products are 3, 7, 6, 2, 3 and 4. The median weight (in kg, up to one decimal place) is _____.

Q3.17.1 NAT 2017 2M

Two machines are defective in a lot of 10. A combination of four machines is to be picked at a time from the lot. The maximum number of combinations that can be obtained without any defective machine is _____.

Q3.16.1 MCQ 2016 1M

A fair coin is tossed N times. The probability that head does not turn up in any of the tosses is

A. $\left(\frac{1}{2}\right)^{N-1}$ **B.** $1 - \left(\frac{1}{2}\right)^{N-1}$ **C.** $\left(\frac{1}{2}\right)^{N}$ **D.** $1 - \left(\frac{1}{2}\right)^{N}$

Q3.16.2 MCQ 2016 1M

A normal random variable X has the following probability density function

$$f_X(x) = \frac{1}{\sqrt{8\pi}} e^{-\left\{\frac{(x-1)^2}{8}\right\}}, \qquad -\infty < x < \infty$$

Then $\int_1^{\infty} f_X(x)\, dx =$

A. 0 **B.** $\frac{1}{2}$ **C.** $1 - \frac{1}{e}$ **D.** 1

Q3.15.1 MCQ 2015 1M

A product is an assembly of 5 different components. The product can be sequentially assembled in two possible ways. If the 5 components are placed in a box and these are drawn at random from the box, then the probability of getting a correct sequence is

A. $\frac{2}{5!}$ **B.** $\frac{2}{5}$ **C.** $\frac{2}{(5-2)!}$ **D.** $\frac{2}{(5-3)!}$

Q3.14.1 MCQ 2014 2M

Marks obtained by 100 students in an examination are given in the table.

Sl. No.	Marks Obtained	Number of Students
1	25	20
2	30	20
3	35	40
4	40	20

What would be the mean, median, and mode of the marks obtained by the students?

A. Mean 33; Median 35; Mode 40

B. Mean 35; Median 32.5; Mode 40

C. Mean 33; Median 35; Mode 35

D. Mean 35; Median 32.5; Mode 35

Q3.14.2 MCQ 2014 2M

In a given day in the rainy season, it may rain 70% of the time. If it rains, chance that a village fair will make a loss on that day is 80%. However, if it does not rain, chance that the fair will make a loss on that day is only 10%. If the fair has not made a loss on a given day in the rainy season, what is the probability that it has not rained on that day?

A. $\frac{3}{10}$ **B.** $\frac{9}{11}$ **C.** $\frac{14}{17}$ **D.** $\frac{27}{41}$

Q3.13.1 MCQ 2013 1M

Customers arrive at a ticket counter at a rate of 50 per *hr* and tickets are issued in the order of their arrival. The average time taken for issuing a ticket is 1 *min*. Assuming that customer arrivals form a Poisson process and service times are exponentially distributed, the average waiting time in queue in *min* is

A. 3 **B.** 4 **C.** 5 **D.** 6

Q3.13.2 MCQ 2013 1M

Let X be a normal random variable with mean 1 and variance 4. The probability $P\{X < 0\}$ is

A. 0.5

B. greater than zero and less than 0.5

C. greater than 0.5 and less than 1.0

D. 1.0

Q3.13.3 MCQ 2013 2M

The probability that a student knows the correct answer to a multiple choice question is $\frac{2}{3}$. If the student does not know the answer, then the student guesses the answer. The probability of the guessed answer being correct is $\frac{1}{4}$. Given that the student has answered the question correctly, the conditional probability that the student knows the correct answer is

A. $\frac{2}{3}$ **B.** $\frac{3}{4}$ **C.** $\frac{5}{6}$ **D.** $\frac{8}{9}$

Q3.13.4 MCQ 2013 1M

A company manufactures 1000 toys every day. On an average, 10% of the toys are defective and 40% of the defective toys can be reworked into defect-free ones. The average number of defect-free toys manufactured daily is

A. 900 **B.** 920 **C.** 940 **D.** 960

Q3.12.1 MCQ 2012 2M

A box contains 4 red balls and 6 black balls. Three balls are selected randomly from the box one after another, without replacement. The probability that the selected set contains one red ball and two black balls is

A. $\frac{1}{20}$ **B.** $\frac{1}{12}$ **C.** $\frac{3}{10}$ **D.** $\frac{1}{2}$

Q3.11.1 MCQ 2011 1M

It is estimated that the average number of events during a year is three. What is the probability of occurrence of not more than two events over a two-year duration? Assume that the number of events follows a Poisson distribution.

A. 0.052 **B.** 0.062 **C.** 0.072 **D.** 0.082

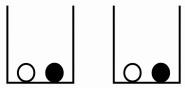
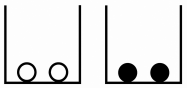
Q3.10.1 MCQ 2010 1M

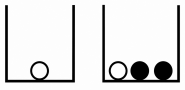
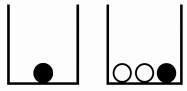
If a random variable X satisfies the Poisson's distribution with a mean value of 2, then the probability that $X \geq 2$ is

A. $2e^{-2}$ **B.** $1 - 2e^{-2}$ **C.** $3e^{-2}$ **D.** $1 - 3e^{-2}$

Q3.10.2 MCQ 2010 2M

Two white and two black balls, kept in two bins, are arranged in four ways as shown below. In each arrangement, a bin has to be chosen randomly and only one ball needs to be picked randomly from the chosen bin. Which one of the following arrangements has the highest probability for getting a white ball picked?

A.  **B.** 

C.  **D.** 

Q3.09.1 MCQ 2009 2M

Consider the joint probability mass function of random variables X and Y as shown in table below: For instance, $P\{X = 1, Y = 2\} = 0.3$

	$X = 1$	$X = 2$
$Y = 1$	0.2	0.3
$Y = 2$	0.3	0.1
$Y = 3$	0.1	0

The value of $P\{X = 2 \mid Y = 2\}$ is

A. 0.10 **B.** 0.25 **C.** 0.40 **D.** 0.75

Q3.08.1 MCQ 2008 1M

For a random variable x $(-\infty < x < \infty)$ following normal distribution, the mean is $\mu = 100$. If the probability is $P = \alpha$ for $x \geq 110$, then the probability of x lying between 90 and 110, i.e., $P(90 \leq x \leq 110)$ will be equal to

A. $1 - 2\alpha$ **B.** $1 - \alpha$ **C.** $1 - \frac{\alpha}{2}$ **D.** 2α

Q3.08.2 MCQ 2008 2M

In a game, two players X and Y toss a coin alternately. Whosoever gets a 'head' first, wins the game and the game is terminated. Assuming that player X starts the game, the probability of player X winning the game is

A. $\frac{1}{3}$ **B.** $\frac{1}{2}$ **C.** $\frac{2}{3}$ **D.** $\frac{3}{4}$

Q3.08.3 MCQ 2008 2M

Customers arrive at a service counter manned by a single person according to a Poisson distribution with a mean arrival rate of 30 per hour. The time required to serve a customer follows an exponential distribution with a mean of 100 seconds. The average waiting time (in hours) of a customer in the system will be

A. 0.138 **B.** 0.166 **C.** 0.276 **D.** 0.332

Q3.07.1 MCQ 2007 1M

Two cards are drawn at random in succession, with replacement, from a deck of 52 well shuffled cards. Probability of getting both 'Aces' is

A. $\frac{1}{169}$ **B.** $\frac{2}{169}$ **C.** $\frac{1}{13}$ **D.** $\frac{2}{13}$

Q3.07.2 MCQ 2007 2M

If X is a continuous random variable whose probability density function is given by

$$f(x) = \begin{cases} K(5x - 2x^2), & 0 \leq x \leq 2, \\ 0, & \text{otherwise} \end{cases}$$

Then $P(X > 1)$ is:

A. $\frac{3}{14}$ **B.** $\frac{4}{5}$ **C.** $\frac{14}{17}$ **D.** $\frac{17}{28}$

Q3.07.3 MCQ 2007 2M

The random variable X takes on the values 1, 2, or 3 with probabilities $(2 + 5P)/5$, $(1 + 3P)/5$, and $(1.5 + 2P)/5$, respectively. The values of P and $E[X]$ are respectively

A. 0.05, 1.87 **B.** 1.90, 5.87

C. 0.05, 1.10 **D.** 0.25, 1.40

Q3.07.4 MCQ 2007 2M

$X_1, \ldots, X_{100}$ are Bernoulli random variables with a probability of success equal to 0.6. By the Central Limit Theorem, the random variable

$$Y = \sum_{i=1,\ldots,100} X_i$$

is approximately normally distributed. Then Y has mean and variance respectively equal to

A. 40 and 24 **B.** 60 and 24

C. 40 and 12 **D.** 60 and 12

Q3.07.5 MCQ 2007 2M

The average number of accidents occurring monthly on an assembly shop floor is 2. The probability that there will be at least one accident in this month is estimated to be

A. 0.055 **B.** 0.456 **C.** 0.865 **D.** 0.950

Q3.05.1 MCQ 2005 1M

Two dice are thrown simultaneously. The probability that the sum of numbers on both exceeds 8 is,

A. $\frac{4}{36}$ **B.** $\frac{7}{36}$ **C.** $\frac{9}{36}$ **D.** $\frac{10}{36}$

Q3.05.2 MCQ 2005 2M

The life of a bulb (in hours) is a random variable with an exponential distribution

$$f(t) = \alpha e^{-\alpha t}, \qquad 0 \le t \le \infty.$$

The probability that its value lies between 100 and 200 hours is

A. $e^{-100\alpha} - e^{-200\alpha}$ **B.** $e^{-100} - e^{-200}$

C. $e^{-100\alpha} + e^{-200\alpha}$ **D.** $e^{-200\alpha} - e^{-100\alpha}$

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Answer Key

ELECTRONICS & COMMUNICATION ENGINEERING					
Q3.26.1 A	Q3.25.1 A	Q3.25.2 7.00	Q3.24.1 0.5	Q3.22.1 4.0	Q3.21.1 B
Q3.21.2 B	Q3.20.1 0.25	Q3.20.2 0.299 to 0.301	Q3.19.1 11	Q3.19.2 0.36 to 0.38	Q3.18.1 0.25
Q3.17.1 0.026 to 0.028	Q3.17.2 2.4 to 2.6	Q3.16.1 0.9 to 1.1	Q3.16.2 0.32 to 0.34	Q3.16.3 0.07 to 0.08	Q3.15.1 C
Q3.15.2 6	Q3.15.3 D	Q3.15.4 6	Q3.15.5 6	Q3.14.1 0.65 to 0.68	Q3.14.2 B
Q3.14.3 49.9 to 50.1	Q3.14.4 C	Q3.14.5 2.9 to 3.1	Q3.14.6 0.15 to 0.18	Q3.14.7 0.79 to 0.81	Q3.14.8 A
Q3.14.9 0.43 to 0.45	Q3.13.1 B	Q3.13.2 D	Q3.12.1 B	Q3.12.2 C	Q3.11.1 C
Q3.10.1 D	Q3.09.1 C	Q3.09.2 C	Q3.09.3 B	Q3.08.1 A	Q3.07.1 A
Q3.07.2 C	Q3.06.1 C	Q3.06.2 D	Q3.06.3 B	Q3.05.1 D	Q3.04.1 D
Q3.03.1 C	Q3.01.1 C	Q3.93.1 0.125	Q3.92.1 B	Q3.88.1 A,B	
ELECTRICAL ENGINEERING					
Q3.26.1 0.1	Q3.25.1 A	Q3.25.2 B	Q3.24.1 B,D	Q3.23.1 0.188	Q3.23.2 C
Q3.22.1 A	Q3.22.2 1	Q3.21.1 B	Q3.19.1 0.26	Q3.19.2 B	Q3.17.1 C
Q3.17.2 A	Q3.17.3 0.9	Q3.16.1 A	Q3.16.2 0.2	Q3.15.1 D	Q3.15.2 D
Q3.15.3 0.25	Q3.14.1 0.42	Q3.14.2 0.4	Q3.14.3 0.142	Q3.14.4 B	Q3.13.1 A
Q3.12.1 C	Q3.12.2 B	Q3.11.1 A	Q3.10.1 C	Q3.09.1 B	Q3.08.1 C
Q3.07.1 C	Q3.06.1 C	Q3.05.1 D	Q3.05.2 B		
INSTRUMENTATION ENGINEERING					
Q3.26.1 1	Q3.26.2 C	Q3.25.1 0.74	Q3.24.1 0.5	Q3.24.2 0.135	Q3.23.1 -0.5
Q3.23.2 240	Q3.22.1 0.33	Q3.20.1 0.09	Q3.20.2 0.7	Q3.19.1 A	Q3.19.2 A
Q3.18.1 D	Q3.18.2 B	Q3.18.3 B	Q3.17.1 C	Q3.16.1 A	Q3.15.1 B

Q3.15.2 4	Q3.14.1 0.890 to 0.899	Q3.14.2 2	Q3.13.1 A	Q3.12.1 B	Q3.11.1 D
Q3.10.1 C	Q3.09.1 C	Q3.08.1 A	Q3.08.2 B	Q3.07.1 A	Q3.06.1 C
Q3.06.2 A	Q3.06.3 A	Q3.05.1 D			
COMPUTER SCIENCE & IT					
Q3.26.1 C	Q3.26.2 B	Q3.26.3 4.24 to 4.26	Q3.26.4 A	Q3.26.5 B	Q3.26.6 0.5
Q3.25.1 0.5	Q3.25.2 0.70 to 0.80	Q3.25.3 0.49 to 0.51	Q3.25.4 0.300 to 0.302	Q3.24.1 A	Q3.24.2 D
Q3.24.3 0.370 to 0.380	Q3.24.4 A,C	Q3.24.5 B	Q3.23.1 C,D	Q3.21.1 B	Q3.21.2 A
Q3.21.3 D	Q3.21.4 15 to 16	Q3.21.5 0.04	Q3.20.1 0.5	Q3.19.1 0.502 to 0.504	Q3.19.2 0.8
Q3.18.1 0.021 to 0.024	Q3.18.2 0.60 to 0.62	Q3.17.1 54	Q3.17.2 0	Q3.17.3 C	Q3.16.1 0.5
Q3.16.2 0.33 to 0.34	Q3.16.3 0.55	Q3.14.1 11.85 to 11.95	Q3.14.2 0.25	Q3.14.3 0.259 to 0.261	Q3.14.4 10
Q3.14.5 3.8 to 3.9	Q3.14.6 0.24 to 0.27	Q3.13.1 C	Q3.12.1 C	Q3.12.2 B	Q3.11.1 C
Q3.11.2 D	Q3.11.3 A	Q3.10.1 A	Q3.09.1 B	Q3.09.2 A	Q3.08.1 A
Q3.08.2 C	Q3.07.1 B	Q3.06.1 A	Q3.05.1 C	Q3.04.1 D	Q3.04.2 D
Q3.04.3 A	Q3.04.4 A	Q3.03.1 C	Q3.03.2 D		
MECHANICAL ENGINEERING					
Q3.26.1 A	Q3.25.1 D	Q3.24.1 0.38 to 0.39	Q3.23.1 A	Q3.22.1 0.17 to 0.19	Q3.21.1 C
Q3.20.1 0.041 to 0.043	Q3.20.2 0.90 to 0.95	Q3.20.3 -1 to 0.98	Q3.19.1 A	Q3.19.2 3	Q3.19.3 0.16 to 0.18
Q3.18.1 B	Q3.18.2 C	Q3.18.3 B	Q3.18.4 A	Q3.17.1 0.75	Q3.17.2 C
Q3.17.3 3.5	Q3.16.1 A	Q3.16.2 0.41	Q3.16.3 99.73	Q3.16.4 A	Q3.15.1 B
Q3.15.2 C	Q3.15.3 D	Q3.15.4 0.97	Q3.15.5 B	Q3.14.1 D	Q3.14.2 A
Q3.14.3 0.26	Q3.14.4 B	Q3.14.5 A	Q3.14.6 0.65	Q3.14.7 74.55	Q3.14.8 50
Q3.14.9 0.8145	Q3.13.1 B	Q3.13.2 D	Q3.13.3 D	Q3.12.1 D	Q3.12.2 B

Q3.11.1 D	Q3.10.1 C	Q3.09.1 D	Q3.09.2 A	Q3.08.1 A	Q3.07.1 D
Q3.06.1 D	Q3.06.2 B	Q3.05.1 B	Q3.05.2 D	Q3.04.1 C	Q3.04.2 D
Q3.03.1 D	Q3.03.2 D	Q3.02.1 A	Q3.02.2 D	Q3.02.3 C	Q3.01.1 D
Q3.00.1 D	Q3.00.2 A	Q3.99.1 C	Q3.98.1 B	Q3.97.1 C	Q3.96.1 D
Q3.93.1 A					
CIVIL ENGINEERING					
Q3.26.1 0.50 to 0.66	Q3.26.2 170.00 to 173.00	Q3.26.3 45.00 to 55.00	Q3.25.1 61.0 to 63.0	Q3.25.2 A	Q3.25.3 B,C
Q3.25.4 2.7 to 2.9	Q3.24.1 C	Q3.24.2 A	Q3.23.1 B	Q3.23.2 0.9	Q3.23.3 B
Q3.22.1 0.78 to 0.82	Q3.22.2 0.018 to 0.020	Q3.22.3 1.9	Q3.21.1 0.58 to 0.60	Q3.21.2 B	Q3.20.1 0.59 to 0.61
Q3.20.2 0.196	Q3.19.1 B	Q3.19.2 0.17 to 0.19	Q3.18.1 0.5	Q3.18.2 C	Q3.18.3 A
Q3.18.4 0	Q3.17.1 209 to 219	Q3.17.2 0.5	Q3.17.3 B	Q3.17.4 0.1276 to 0.1372	Q3.16.1 B
Q3.16.2 A	Q3.16.3 A	Q3.16.4 54.49 to 54.51	Q3.15.1 0.23 to 0.25	Q3.15.2 1.06 to 1.07	Q3.14.1 0.4
Q3.14.2 0.25 to 0.28	Q3.14.3 B	Q3.14.4 0.26 to 0.27	Q3.14.5 B	Q3.13.1 D	Q3.13.2 6
Q3.12.1 B	Q3.12.2 D	Q3.12.3 A	Q3.11.1 C	Q3.11.2 D	Q3.10.1 C
Q3.09.1 D	Q3.08.1 A	Q3.08.2 B	Q3.08.3 D	Q3.07.1 C	Q3.06.1 B
Q3.06.2 D	Q3.05.1 D	Q3.04.1 C	Q3.03.1 C		
CHEMICAL ENGINEERING					
Q3.26.1 0.5	Q3.25.1 D	Q3.25.2 A	Q3.24.1 B	Q3.23.1 A	Q3.21.1 0.069 to 0.071
Q3.20.1 B	Q3.19.1 0.24 to 0.26	Q3.18.1 C	Q3.17.1 C	Q3.17.2 0.35 to 0.37	Q3.16.1 2.25 to 2.72
Q3.14.1 6.99 to 7.01	Q3.14.2 C	Q3.13.1 0.083	Q3.13.2 B	Q3.12.1 C	Q3.10.1 B
Q3.10.2 B	Q3.09.1 A	Q3.08.1 B	Q3.06.1 B,C	Q3.06.2 D	Q3.05.1 C

Q3.04.1 B	Q3.02.1 D				
PRODUCTION & INDUSTRIAL ENGINEERING					
Q3.26.1 0.78 to 0.82	Q3.26.2 B	Q3.26.3 0.14 to 0.16	Q3.25.1 B	Q3.24.1 50	Q3.24.2 4
Q3.23.1 B	Q3.23.2 A	Q3.22.1 0.13 to 0.15	Q3.22.2 0.70	Q3.21.1 2	Q3.21.2 0.365 to 0.370
Q3.20.1 C	Q3.20.2 3	Q3.19.1 0.59 to 0.61	Q3.18.1 0.49 to 0.51	Q3.18.2 3.49 to 3.51	Q3.17.1 70
Q3.16.1 C	Q3.16.2 B	Q3.15.1 A	Q3.14.1 C	Q3.14.2 D	Q3.13.1 C
Q3.13.2 B	Q3.13.3 D	Q3.13.4 C	Q3.12.1 D	Q3.11.1 B	Q3.10.1 D
Q3.10.2 C	Q3.09.1 B	Q3.08.1 A	Q3.08.2 C	Q3.08.3 B	Q3.07.1 A
Q3.07.2 D	Q3.07.3 A	Q3.07.4 B	Q3.07.5 C	Q3.05.1 D	Q3.05.2 A

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